Enhancing Revenue Collection Capacity through Automation: Evidence from a VAT*

Bhanu Gupta† Tejaswi Velayudhan‡ Vedant Vohra§ Ankur Garg¶

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Abstract

Tax administrations worldwide are investing heavily in digitization, creating new enforcement opportunities. Can low-cost enforcement tools built atop this digital infrastructure increase tax capacity without adverse consequences? We study an Indian intervention implementing automated cross-checks to target fraudulent input tax credits. Using administrative data, we show that the policy reduced excess credits and increased tax payments, reducing evasion via excess credits by 1.2-4.3%. The resulting revenue gains likely exceed revenue losses from the policy's blunt targeting. This type of policy is readily implementable in many developing countries that already have digital filing but are yet to exploit its full potential.

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†Department of Economics, Ashoka University. Email: bhanu.gupta@ashoka.edu.in
‡Department of Economics, University of California, Irvine. Email: tvelayud@uci.edu
§Department of Economics, University of California, San Diego. Email: vohra@ucsd.edu
¶Department of Trade and Taxes, Government of NCTD.

1 Introduction

Information is central to a state's capacity to enforce a tax system.[1] Digitization of tax administration has enabled both the collection and use of vast troves of information in developed and developing countries alike. This digital architecture interacts particularly meaningfully with the Value Added Tax (VAT), whose main advantage over a retail sales tax lies in the information generated. Each transaction between firms generates dual reports—sellers report the transaction as a sale while buyers report it as a purchase—creating independently verifiable information for the tax authority. Although digitization has spread widely—for example, at least 70% of countries across income levels mandate electronic filing of VAT returns—the use of these data for enforcement still lags. Fewer than 30% of low-income countries perform any automated cross-checks.[2] This paper demonstrates the potential of such automated checks to curb evasion with minimal additional investment, in the context of India's Goods and Services Tax (GST).

A key vulnerability of VAT systems is the provision of fraudulent input tax credits (ITCs): tax-registered sellers provide invoices that generate tax credits for their clients without remitting taxes on those sales. Fraudulent credits erode 2–4 percent of net VAT collections in low-income countries (about 3 percent in India), with marked spillovers along production chains (Carrillo et al., 2017; Waseem, 2023; Almunia et al., 2024).[3] Beginning in August 2021, the online GST filing portal automatically blocked firms from passing on new ITC to clients if they had not filed two of their payment-associated tax returns, meaning they had made no payment for two months. The reform neither raised existing statutory penalties nor required intervention by tax personnel; instead, it harnessed the existing e-filing infrastructure to automate enforcement.

While this type of reform is feasible in most developing countries with e-filing mandates, adoption remains low. One barrier to adoption may be the concern that such an automated rule is a blunt instrument: evaders might easily circumvent it, while legitimate taxpayers may bear unintended costs. The reform conditions new credit issuance on the timely filing of payment-linked returns, encouraging firms to report and make tax payments, or else blocking them from issuing new credits. The rule is triggered by verifiable, past non-compliance, but can result in blocking of *future* tax credits, which may or may not be legitimate. The success of the policy therefore depends on the extent to which it can reduce evasion while imposing minimal cost on tax revenue-generating activity. We provide quantitative evidence on this tradeoff. We show that the policy meaningfully reduced targeted evasion, yielding revenue gains commensurate with its low administrative cost. However, the policy leaves a large share

[1] A large literature has shown the empirical and theoretical relevance of information for tax capacity (Kleven et al., 2016; Kumler et al., 2020; Jensen, 2022; Pomeranz, 2015; Best et al., 2021; Kleven et al., 2011).

[2] Authors' calculations using IMF RA-FIT 2022 data (https://data.rafit.org, accessed April 1, 2025).

[3] Evasion and revenue estimates are reported in Lok Sabha Questions (2024) and Goods and Services Tax Network (2023), respectively.

of excess credit issuance unchecked and blocks some taxpayers who would otherwise have generated revenue.

To quantify these effects, we leverage administrative tax data from Delhi. Using these data, we identify taxpayers who generated potentially fraudulent credits in the pre-policy period, classifying them as Fake Credit Generators (FCGs). Despite comprising only 2.8 percent of continuously-registered taxpayers, these FCGs account for roughly 84 percent of excess credits in the pre-policy period and supply these credits to 3.4 percent of firms.

Targeted taxpayers could respond to the policy by reducing their delinquency on payment-linked returns or by ceasing tax credit issuance. Using an event-study design, we find that FCGs start issuing fewer credits. Compared to non-FCGs with otherwise similar characteristics, FCGs become 3 percentage points less likely to issue credits to their clients. Part of this decline is mechanical: the rule prevents FCGs with overdue payment-linked returns from issuing credits. We also find evidence of deterrence: even FCGs unlikely to be blocked issue fewer credits, consistent with a higher perceived risk of enforcement. Furthermore, FCGs' payments do not increase and there is no evidence that firms try to circumvent the rule by filing returns with only token payments. Instead, FCGs become 1.5 percentage points more likely to become inactive by not filing both returns and risking subsequent deregistration.

The effects of this disruption propagate downstream: fewer credits issued by FCGs mean that their clients have fewer credits available to offset their gross tax liabilities. We examine whether these clients replace the lost credits, and if not, whether they increase net tax payments or reduce their reported gross tax liabilities. We find that client firms experience a decline in credits and report an imprecisely estimated decline in gross payments. This results in a lasting decrease in the share of their tax liability paid using ITCs and an increase in net tax payments. Aggregating the decline in credit issuance across all FCGs, the policy generated annual revenue gains between INR 1.5 billion and 5.2 billion—which is 1.2 to 4.3 percent of all suspected ITC evasion, or 0.34 to 1.20 percent of net GST collections in Delhi—at negligible additional administrative cost.[4]

One limitation of our analysis is that excess credits may be generated not only by wholly fraudulent firms but also by firms with some legitimate economic activity. For example, firms may be temporarily delinquent in their tax payments but issue tax credits. If the policy causes such firms to shut down, it results in a loss of tax revenue and a decrease in real economic activity. A full understanding of the policy's welfare effects would require us to measure both. However, our data only allow us to measure the effect on revenue, leaving out any effect on production that is not also reflected in revenue. A descriptive simulation suggests that the revenue lost from legitimate firms could be a quarter of the revenue gained from curbing excess credits.

Our work shows the promise of a low-cost intervention to target excess credits, contrasting

[4] 1 USD equaled INR 74 in 2021. Net GST collections in Delhi were INR 435.5 billion in FY 2021–22 (Goods and Services Tax Network, 2023).

with more resource-intensive audits (Carrillo et al., 2023; Waseem, 2023) and algorithm-based risk scoring (Mittal et al., 2018; Shah, 2026). These tools have not realized their full potential in developing countries due to factors ranging from institutional frictions that hinder the integration of sophisticated algorithms (Barwahwala et al., 2024) to limited state capacity to recover known evasion (Okunogbe and Tourek, 2024; Best et al., 2021). In this study, we show that enforcement tools at the pre-filing stage can effectively curb evasion without relying on the state's capacity to recover lost revenue post-filing. While other strategies to target excess credits can yield larger gains, they are also costlier to implement. For instance, notifying clients of fake firms—identified after an intensive and costly "antighost" initiative—increased tax payments by 13.7 percent among treated firms in Ecuador (Carrillo et al., 2023). In Pakistan, the use of sophisticated software that verifies the authenticity of each invoice increased net tax collections by 10 percent (Shah, 2026). Our estimates indicate that meaningful gains can be made simply by building on existing digital filing architecture.

Our findings also highlight a technology that replaces, rather than augments, the work of tax officials. Many interventions increase the productivity of state officials by helping them assess tax liability or detect evasion, but they ultimately still require human action to collect the taxes owed. For instance, high-tech tools are used to assess property taxes in several contexts (Dzansi et al., 2022; Casaburi and Troiano, 2016; Meiselman, 2018), but personnel must still deliver tax bills and pursue unpaid liabilities. Similarly, e-filing, e-invoicing, and electronic billing machines provide valuable information for enforcement (Bellon et al., 2022; Okunogbe and Pouliquen, 2022), yet converting this data into revenue remains resource-intensive. In contrast, the policy we study obviates the need for manual intervention in enforcement and collection, speaking to the benefits of automation over discretion (Bachas et al., 2025). We evaluate its effectiveness in targeting evasion alongside its potential unintended consequences for taxpayers, a prevalent concern regarding enforcement efforts in developing countries (Almunia et al., 2024; Mascagni et al., 2023; Brockmeyer et al., 2024).

2 Institutional Context

2.1 Tax Filing and Payment in the GST

The Indian Goods and Services Tax (GST) is a nationwide value-added tax: firms remit taxes on their sales net of credits received for tax remitted on their inputs. Taxpayers self-report their total gross tax liability on the "GSTR-3B" return (henceforth, the *payment-linked* return). They must file this return and make their payment at the same time. The payment amount is net of their available input tax credits (ITCs). Firms also report invoice-level sales in the "GSTR-1" return, which we refer to as the *credit-linked* return since it generates ITC for GST-registered client firms. ITC available to such clients is limited to what a seller has reported on this return,

which automatically populates a "GSTR-2" return visible to the client.

In theory, total gross tax liability reported on the payment-linked return must either match or exceed the tax credits issued on the credit-linked return, since the latter excludes taxable sales to non tax-registered entities like final consumers. But in practice, there are discrepancies. Sellers can fraudulently provide credits without paying associated taxes by reporting higher sales in their credit-linked return than in their payment-linked return. We term such credits as "excess credits".[5]

2.2 Policy Change: Automated Blocking of Credit Provision

To curb the provision of excess credits, a nationwide GST reform in August 2021 automatically blocked firms from filing their credit-linked return if they had not filed a payment-linked return for the preceding two periods.[6] Any firm affected by this policy could respond in two ways: (1) file any pending payment-linked returns and continue to file returns with no more than a 2-month delay going forward, or (2) stop filing any credit-linked returns. Note that firms that comply with the policy could still continue to generate excess credits by under-reporting sales on the payment-linked returns compared to what they report on their credit-linked return. They could simply file the payment-linked return with a nominal or zero tax payment (a "nil" return) and leave other behavior unchanged. We test for each of these potential responses in Section 5.1.

3 Data

We use administrative data from the Delhi Department of Trade and Taxes covering the universe of GST taxpayers registered in Delhi between January 2021 and April 2022. For each firm and return period, we observe whether and when each return was filed, gross and net tax payments reported in the payment-linked return, and input tax credits issued through the credit-linked return. We compare gross tax payments with credits issued to tax-registered buyers to identify potential Fake Credit Generators (FCGs), as described in Section 3.1. Invoice-level sales reported in the credit-linked return also allow us to construct firm-to-firm trade linkages and trace how the policy's effects propagate through supply chains. Registration records provide additional firm characteristics, including sector (e.g., retail, manufacturing) and firm age. We received registration records as monthly snapshots, except for February–June 2021 and August 2021, so we do not observe the exact month of deregistration for firms deregistered in those months.

[5] In standard VAT systems, an equivalent behavior would be issuing tax invoices to buyers but failing to remit taxes on those sales.

[6] Appendix Figure A.1 confirms that the policy was effectively implemented: the share of firms non-compliant with the blocking rule is zero after implementation.

Our main sample consists of firms that are required to file GST returns on a monthly basis and were registered for all return periods from January 2021 to April 2022, covering seven months before and nine months after the introduction of the blocking policy in August 2021. This restriction allows us to examine firms' filing behavior without conflating it with changes in the sample caused by tax authority-initiated deregistrations. Deregistration may be an outcome of the policy, but in our setting it is rarely initiated by the taxpayer: nearly all deregistrations are initiated by the tax authority. Taxpayers intending to deregister would instead simply stop filing both returns and become inactive. Therefore, a key outcome of interest in our setting is whether a firm becomes inactive, which we can directly and reliably observe.

We repeat our main causal analysis on an expanded sample of firms that were registered for at least one period between January and June 2021 in Appendix Section B and discuss their contribution to the aggregate revenue effects of the policy. This appendix also separately analyzes smaller firms (with annual taxable revenue below INR 50 million) that elect to file returns quarterly, because the policy permits a one-quarter grace period for these firms rather than two months.

3.1 Identifying Fake Credit Generators from Tax Returns

Excess credits arise when firms issue input tax credits that are not backed by corresponding tax payments.[7] Under the GST, this occurs when a seller reports sales in the credit-linked return but fails to remit payment on those sales in the payment-linked return, resulting in a positive discrepancy that we can observe in our data by comparing the two returns. Our target is to identify how firms that generate excess credits in the pre-policy period respond to the policy.

Between January and June 2021, around 9% of firms reported higher sales in their credit-linked return than in their payment-linked return. Although all of these discrepancies represent evaded or delayed payment regardless of whether the underlying transaction or firm was real or fake, some could be attributed to mistakes. Therefore, we classify firms as Fake Credit Generators (FCGs) using two mutually exclusive criteria.

First, among firms with positive tax payments between January and June 2021, we classify those whose total credit issued is at least 10% larger than their total tax payments as FCGs, yielding 5,644 firms (2.7% of all firms). Second, among firms that make no tax payments between January and June 2021 but still issue tax credits to their buyers, we also classify as FCGs those with total credit provision above the 75th percentile, yielding an additional 167 firms. These two criteria result in a total of 5,811 FCGs, which together account for 84% of all excess credit issued during this six-month period.

All taxpayers were subject to the blocking policy at the same time, leaving no control group

[7] Firms could also misreport the identity of their clients, transferring invoices to taxpayers like exporters who might benefit more from the credits (Waseem, 2023). Because this form of evasion cannot be identified from tax returns alone, we focus on cases in which reported credits exceed reported payments.

exempt from the reform. We use pre-policy FCG status to identify firms that were more likely to be mechanically blocked or to change their behavior in response to a higher perceived risk of enforcement. FCG status need not coincide with actual blocking: some FCGs may never be blocked, while some non-FCGs may also be affected. Nonetheless, FCGs are at greatest risk of being affected by the blocking rule and they exhibit the evasion behavior (issuing excess credits) that the reform seeks to curb.[8]

To explore the mechanical versus deterrence effects of the blocking rule, we divide FCGs into two categories based on their pre-policy filing behavior. *Filer FCGs* filed all their payment-linked returns between January and June 2021 but reported substantially higher sales in their credit-linked returns; they account for 5,211 of the 5,811 FCGs. Because the rule requires two consecutive unfiled payment returns, an immediate response by these firms in the post-policy period would be consistent with a deterrence channel through increased perceived enforcement. In contrast, firms that failed to file at least one payment-linked return during this period are defined as *Non-Filer FCGs*; they account for 600 of the 5,811 FCGs.[9] Their prior non-filing behavior suggests that they are more likely to be mechanically blocked from filing their credit-linked return by the policy.

There are three main reasons that excess credits could arise. First, there may be collusive fraud between sellers and buyers to issue invoices without any underlying real transaction. Second, sellers could be making legitimate sales but evading taxes while issuing credits. Third, sellers may experience a temporary shock causing them to delay payments but issue credits on time. In the case of collusive fraud, we expect that insofar as the policy is effective, it will result in a pure transfer from firms to the government. In the latter two cases, the policy will result in a higher effective tax rate, which could lead sellers or their client firms (who benefit from the credits) to produce less output or even close altogether. Our data and design can measure changes to output reported in tax data for both sellers and client firms, which is a combination of reporting and real effects. But because of lack of data on firms' real outcomes, we cannot distinguish between wholly fraudulent firms and those with some real activity. Any impact on production of the latter, which was not reported in tax data, is unmeasured. For example, for firms that were evading payment on only part of their output, we cannot measure any change to the unreported portion of their output. Any such adverse unintended outcomes are only measured to the extent they are reflected in tax revenue.

[8]FCGs are more likely than non-FCGs to file credit-linked returns while leaving payment-linked returns unfiled, confirming greater ex ante exposure to the blocking rule as the classification intends (Appendix Figure A.1). FCGs also issued more credits, reported lower gross tax liabilities, and were more likely to operate in the services sector (Appendix Table A.1).

[9]Notably, the number of Non-Filer FCGs is comparable to the 800 ghost firms identified through substantial tax authority effort in Ecuador (Carrillo et al., 2023).

4 Empirical Strategy

To identify how the blocking policy affected firms issuing excess credits in the pre-policy period, we compare changes in filing behavior between FCGs and non-FCGs in an event-study framework. Because the policy applied to all firms, the estimates capture FCGs' response relative to that of non-FCGs. To construct a comparable control group, we use Coarsened Exact Matching (CEM) to match FCGs with non-FCGs on pre-policy characteristics: firm age, firm size, nature of business, whether the firm reports inter-state sales, and the number of payment returns filed before the policy (Blackwell et al., 2009). Our preferred specification in Table 1 uses CEM weights; for robustness, we also report estimates using the full unmatched control group in Figure 1.

Formally, we estimate the following two-way fixed effects event-study specification:

$$y_{it} = \gamma_i + \lambda_t + \sum_{\substack{k=-7 \\ k\neq -2}}^{8} \beta_k \times Treated_i \times \mathbb{1}(t = k) + \varepsilon_{it}, \tag{1}$$

where y_{it} denotes the outcome for firm i in return period t, with $t = 0$ corresponding to August 2021; γ_i are firm fixed effects; and λ_t are filing-period fixed effects. June 2021 ($t = -2$) is the omitted period.[10] $Treated_i$ indicates pre-policy FCG status. Each coefficient β_k estimates the change in the average outcome for FCGs from June 2021 to period k, minus the corresponding change for non-FCGs. Standard errors are clustered at the firm level throughout. For the difference-in-differences estimates of filing outcomes, we restrict the sample to return periods through February 2022 because data were obtained from the tax administration in June 2022, and it is possible that some firms filed March–April 2022 returns after the data were collected.

To assess how the policy affected downstream firms, we use invoice-level data to measure their pre-policy exposure to FCG suppliers. We define treated downstream firms as those that received at least 5% of their total input tax credits from FCGs between January and June 2021. The control group consists of non-FCG firms that received no credits from FCGs during this period.[11] We then estimate Equation 1, with this treatment and control group.

Identification relies on the assumption that, absent the policy, outcomes for treated and control firms would have followed parallel trends, and that firms did not alter behavior in anticipation of the reform. The event-study estimates presented in Section 5 show stable pre-trends and no evidence of anticipatory responses, providing support for these assumptions. Under these assumptions, the post-policy coefficients identify the causal effect of the policy

[10] We use June rather than July as the reference period because the policy may have affected the filing of July returns.

[11] To ensure that control firms received no pre-policy credits from any identified FCG, we also exclude clients of FCGs, that were not continuously registered throughout the sample period or elected to file returns quarterly. Firms that received a positive but negligible share (less than 5%) of their total input tax credits from FCGs between January and June 2021 are excluded from this analysis.

on treated firms relative to its effect on control firms. Interpreting these coefficients as the average treatment effect on the treated (ATT) additionally requires that the policy had no effect on control firms on average. Section 5 provides suggestive evidence consistent with limited effects on control firms' filing behavior. A key threat to identification is mean reversion: firms with large pre-policy discrepancies between reported credits and payments may mechanically reduce excess credits over time, independent of the blocking policy. This could generate an apparent post-reform decline in discrepancies even without the policy. The results presented in Section 5 provide evidence against this explanation.

5 Impact of Automated Checks on Excess Credits and Revenue

We evaluate the revenue consequences of the policy in three steps. First, we estimate the causal effect of the policy on the direct compliance margins—specifically, the filing of payment- and credit-linked returns—of FCGs relative to non-FCGs. Second, we trace these effects through the supply chain by examining the responses of the FCGs' downstream clients. Finally, we synthesize these estimates to quantify the net revenue impact of the intervention, weighing the curtailed evasion against the revenue lost from potentially disrupting firms that were only temporarily non-compliant.

5.1 Impact on Targeted Evasion: Provision of Excess Input Tax Credits

Our first key finding is that the policy successfully reduces the provision of excess credits by FCGs. It does so by reducing credit provision rather than increasing tax payments.

We begin with Non-Filer FCGs, the group most exposed to the policy because they failed to file at least one payment-linked return between January and June 2021. Panel A of Table 1 shows they become 11 percentage points less likely to file their credit-linked returns (Column 1) and no more likely to file their payment returns (Column 2). As a result, they are 10 percentage points more likely to become inactive, i.e., stop filing all returns (Column 3). Although Non-Filer FCGs account for 0.3 percent of all firms, they supplied roughly 4 percent of all excess credits in the pre-policy window among our main sample of continuously-registered firms. This share falls to almost zero after the reform (Figure 2), indicating that excess credit issuance by Non-Filer FCGs is nearly eliminated post-policy. These responses may arise from mechanical blocking alone and do not necessarily indicate a change in taxpayer behavior.

Filer FCGs show a smaller response, but one more consistent with a behavioral change due to a higher perceived risk of enforcement. These firms had filed all payment-linked returns for January–June and therefore could not have been mechanically blocked from filing their credit-linked return in August, the first post-policy month. Nevertheless, their credit-linked

filing rate falls immediately by 2 percentage points and remains stable thereafter (Figure 1, Panel A). They also become 0.6 percentage points more likely to become inactive (Table 1, Panel B). After August, Filer FCGs could become subject to mechanical blocking if they stopped filing payment-linked returns. To assess how much blocking contributes to the estimated decline, we construct an alternative outcome that treats every blocked firm as having filed its credit-linked return: only firms that were able to file but chose not to are counted as non-filers. Re-estimating the event-study specification with this outcome produces a nearly identical decline in filing (Appendix Figure A.2), suggesting that most of the estimated response reflects a behavioral change rather than mechanical blocking.

Because Filer FCGs account for 90 percent of all FCGs, the estimates for the full FCG sample closely track their response. Table 1, Panel C shows a 2.9 percentage point decline in credit-linked return filing, a slight decline in payment-linked return filing, and a 1.5 percentage point increase in inactivity.[12] We also find no evidence that FCGs circumvent the policy by filing payment-linked returns with nominal or nil payments (Appendix Figure A.3), suggesting that the policy had a deterrent effect.

The event-study estimates in Panel A of Figure 1 show small pre-policy coefficients that fluctuate around zero with no systematic trend, providing support for our identification assumption.[13] The post-policy drop in filing of the credit-linked return is immediate and persistent.

We provide two further pieces of evidence that the filing responses are policy-induced and concentrated among firms generating excess credits. First, the decline in credit-linked return filing increases with the size of the pre-policy discrepancy between credits issued and corresponding payments. The decline is 6.5 percentage points for firms with discrepancies of at least 50 percent, compared with 4 percentage points for those with discrepancies of at least 10 percent (Figure 1, Panel B). For firms with positive discrepancies below 10 percent, which may reflect reporting errors, the estimated decline is only 0.4 percentage points relative to firms with no discrepancies (Appendix Table A.3). Together, these results suggest that filing responses were concentrated among firms generating substantial excess credits, and are consistent with limited effects on non-FCGs' filing behavior. Second, the results are unlikely to reflect firms already shutting down or mean reversion arising from our pre-policy FCG classification. To assess these possibilities, we classify FCGs using only January–March 2021 and examine their filing behavior between April and the policy's introduction in August. Either explanation would predict declining filing rates during this pre-policy period. However, filing rates fall significantly only after the policy takes effect (Appendix Figure A.7). This suggests

[12] The increase in inactivity is persistent: FCGs become 1.4 percentage points more likely to file neither return for three consecutive months (Appendix Table A.2), close to the 1.5 percentage point effect for a single month (Table 1, Panel C).

[13] The policy was enacted within two months of the July payment deadline; the observed July effect in Figure 1 likely reflects FCGs with pending returns choosing not to file. Event-study estimates for all FCGs' filing of the credit- and payment-linked returns show no pre-trends (Appendix Figures A.4 and A.5), and pre-policy filing rates are similar across groups (Appendix Figure A.6).

that the decline in filing was induced by the policy rather than by classifying firms already winding down operations as FCGs or by mean reversion.

Our main sample consists of firms registered throughout the study period, so that observed changes in filing do not reflect deregistrations initiated by the tax authority. To assess sensitivity to this restriction and capture the policy's broader revenue effects, we expand the sample to include firms registered for at least one month between January and June 2021. In this expanded sample, FCGs become 6 percentage points less likely to file credit-linked returns, 1.7 percentage points less likely to file payment-linked returns, and 5.4 percentage points more likely to become inactive (Appendix Table B.1, Columns 1-3). For this sample, we can also estimate the impact on deregistration itself, which we cannot by construction estimate in our main sample. We find that all FCGs become 1.9 percentage points more likely to be deregistered (Appendix Table B.1, Column 4). This increase in deregistration probability is consistent with the tax authority being more likely to deregister inactive firms.[14]

A potential cost of the policy is that blocking firms with temporary compliance lapses may cause them to become inactive, reducing future tax revenue they would otherwise have generated. We discuss this tradeoff in Section 5.3.

5.2 Downstream Effects on Client Firms

Next, we turn to the downstream effects of this loss of credits on client firms. The reduction in FCG tax credits can elicit three distinct responses from downstream clients. If clients leave their reported output unchanged and do not replace the lost credits with alternative purchases (real or fake), their net tax payments will increase. If they successfully substitute the credits, net payments remain unchanged. If clients decrease their reported output either because of a decrease in real output or through underreporting, net payments could decrease.

We identify clients of FCGs as firms receiving at least 5 percent of their input tax credits from FCGs, comparing them to a control group receiving no FCG credits; FCGs themselves are excluded from this analysis.[15] We report effects on clients of Non-Filer and Filer FCGs in Panels A and B of Table 2, respectively, and on the clients of all FCGs in Panel C. The estimated responses are similar among clients of Non-Filer and Filer FCGs. These firms increase net tax payments as a share of total tax payments by 1.3–1.7 percentage points (Table 2, Column 1). This increase in the share of taxes paid in cash begins immediately after the policy and persists throughout the post-policy period, suggesting that client firms do not fully recover the lost ITC (Figure 3). In contrast, total gross payments decline by a statistically insignificant 1.7-2.2

[14]Because registration data are unavailable for February–June 2021 and August 2021, we do not observe the exact month of deregistration for firms deregistered in those months and cannot test for pre-trends in this outcome. Among firms continuously registered from January to September 2021, deregistration of FCGs rises steadily after the policy. By April 2022, FCGs are about 4 percentage points less likely than non-FCGs to remain registered, and Non-Filer FCGs about 34 percentage points less likely (Appendix Figure A.8).

[15]These client firms average higher gross tax liability but lower ratio of net payments to gross liability (Appendix Table A.4).

percent (Table 2, Column 3).[16,17]

Overall, these findings demonstrate that the policy successfully increased net tax payments from FCG clients. We account for the potential decline in gross tax payments when estimating aggregate revenue effects in the next section.

5.3 Revenue Implications

We first estimate the revenue gain from reduced credit provision by FCGs, net of any accompanying decline in their tax payments. Between January and June 2021, FCGs in our main sample generated approximately INR 4.5 billion in monthly credits.[18] Multiplying this amount by the 2.9 percentage point decline in credit-linked filing (Table 1, Panel C, Column 1) and annualizing implies a decline in credits of INR 1.6 billion. Repeating this calculation for payment-linked filing implies an annual decline in net payments by FCGs of INR 0.06 billion, leaving a net revenue gain of approximately INR 1.5 billion. Including deregistered firms increases this gain to INR 5.2 billion.[19] These gains amount to 1.2% and 4.3%, respectively, of estimated annual ITC evasion (INR 121 billion) in Delhi in FY 2023–2024.[20] Some of these gains could be undone if client firms replace lost credits with other fraudulent credits or reduce their reported gross tax liabilities. Client responses do not appear to offset these gains: clients of continuously-registered FCGs increase their annual net payments by INR 2.2 billion.

Next, we estimate the potential revenue loss from the rule's impact on firms that are only temporarily non-compliant but are blocked from issuing credits. If the inability to issue credits causes these firms to become inactive, the tax authority may lose future revenue they would otherwise have generated. To estimate this loss, we counterfactually apply the policy rule to taxpayers' filing behavior between January and April 2021 and identify firms that would have been blocked. We then ask whether these firms would have been classified as FCGs based on their filing and payment behavior in May and June 2021. We find that the rule would have blocked about 1% of non-FCGs, or 2,142 firms (Appendix Figure A.10). In May and June 2021, these firms contributed net revenue of INR 90.5 million per month. If these taxpayers are 34 percentage points more likely to be deregistered, matching the April 2022 gap in deregistration rates between Non-Filer FCGs and non-FCGs (Appendix Figure A.8), the annualized revenue loss would be INR 369 million, about a quarter of the INR 1.5 billion net revenue gain from FCGs. Applying the policy's causal effect on inactivity among Non-Filer FCGs (9.7 percentage points; Table 1, Panel A, Column 3) implies a revenue loss of INR 105 million, or 7 percent

[16] Our expanded sample shows a decline of 2.7 percent (Appendix Table B.2). We observe no net payment impact on clients of quarterly-filing FCGs, as these smaller firms are less significant sources of fraudulent credits.

[17] After the policy, client firms become 0.4 percentage points less likely to report any sales (Appendix Table A.5), and they are deregistered at rates no higher than those of control firms (Appendix Figure A.9).

[18] Appendix C provides the details of all calculations reported in this paragraph.

[19] Repeating the calculation for quarterly filers adds INR 0.1 billion.

[20] Annualized from the tax authority's estimate of INR 30.3 billion in suspected evasion in Delhi during Q4 2023 (Lok Sabha Questions, 2024).

of the gain. At the other extreme, if all blocked non-FCGs were deregistered, the annualized revenue loss would total INR 1.09 billion, or 72 percent of the gain.

At the same time, the simulated rule would have blocked only 2.6% of FCGs, consistent with most FCGs filing their payment-linked returns. The remaining 97.4% issued INR 7.2 billion in excess credits from August 2021 to April 2022.

Finally, because the policy blocks few FCGs while also blocking some legitimate firms, we ask whether alternative automated triggers could target excess credit issuance more precisely. We contrast a blocking rule based directly on the discrepancy between credits issued and tax payments against blocking rules that allow different grace periods for delinquent payment-linked returns.[21] A rule that blocked firms whose credits issued exceeded their payments by more than 10 percent would have caught 29 percent of FCGs with a false positive rate of 2.6 percent. By contrast, shortening the grace period from two months to one would have produced a similar false positive rate of 2.9 percent but caught only 6 percent of FCGs. These simulations consider only the mechanical effects of alternative reforms and ignore any behavioral responses. With this caveat in mind, they suggest that a discrepancy-based rule could block substantially more FCGs than a shorter grace period without increasing the share of non-FCGs blocked.

6 Conclusion

The digitization of tax administration in developing countries offers untapped potential for automated, low-cost enforcement. We show that simply requiring firms to file their payment-linked returns before issuing credits generates annual revenue gains of INR 1.5–5.2 billion, or 1.2–4.3 percent of suspected evasion via excess credits. Targeted fraudulent credit-generating firms are more likely to become inactive and their clients increase net tax payments. These gains must be weighed against unobserved welfare losses from eliminating a potentially valuable source of flexibility for taxpayers. This blunt rule cannot target all fraudulent credits but meaningfully raises revenue at negligible administrative cost.

[21] Appendix Figure A.10 reports false positive and negative rates for the alternative grace periods and discrepancy thresholds.

References

Almunia, Miguel, Jonas Hjort, Justine Knebelmann, and Lin Tian, "Strategic or confused firms? evidence from "missing" transactions in Uganda," *Review of Economics and Statistics*, 2024, *106* (1), 256–265.

Bachas, Pierre, Anne Brockmeyer, Alipio Ferreira, and Bassirou Sarr, "Algorithms and bureaucrats: evidence from tax audit selection in Senegal," Technical Report, The World Bank 2025.

Barwahwala, Taha, Aprajit Mahajan, Shekhar Mittal, and Ofir Reich, "Is Model Accuracy Enough? A Field Evaluation Of A Machine Learning Model To Catch Bogus Firms," Technical Report, National Bureau of Economic Research 2024.

Bellon, Matthieu, Era Dabla-Norris, Salma Khalid, and Frederico Lima, "Digitalization to improve tax compliance: evidence from VAT e-Invoicing in Peru," *Journal of Public Economics*, 2022, *210*, 104661.

Best, Michael, Jawad Shah, and Mazhar Waseem, "Detection Without Deterrence: Long-Run Effects of Tax Audit on Firm Behavior," 2021.

Blackwell, Matthew, Stefano Iacus, Gary King, and Giuseppe Porro, "cem: Coarsened exact matching in Stata," *The Stata Journal*, 2009, *9* (4), 524–546.

Brockmeyer, Anne, Giulia Mascagni, Vedanth Nair, Mazhar Waseem, and Miguel Almunia, "Does the Value-Added Tax Add Value? Lessons Using Administrative Data from a Diverse Set of Countries," *Journal of Economic Perspectives*, 2024, *38* (1), 107–132.

Carrillo, Paul, Dave Donaldson, Dina Pomeranz, and Monica Singhal, "Ghosting the tax authority: fake firms and tax fraud in Ecuador," *American Economic Review: Insights*, 2023, *5* (4), 427–444.

_ **, Dina Pomeranz, and Monica Singhal**, "Dodging the taxman: Firm misreporting and limits to tax enforcement," *American Economic Journal: Applied Economics*, 2017, *9* (2), 144–164.

Casaburi, Lorenzo and Ugo Troiano, "Ghost-house busters: The electoral response to a large anti–tax evasion program," *The Quarterly Journal of Economics*, 2016, *131* (1), 273–314.

Dzansi, James, Anders Jensen, David Lagakos, and Henry Telli, "Technology and tax capacity: evidence from local governments in Ghana," Technical Report, National Bureau of Economic Research 2022.

Goods and Services Tax Network, "Table: State wise Tax Collection," https://www.gst.gov.in/download/gststatistics 2023.

Jensen, Anders, "Employment structure and the rise of the modern tax system," *American Economic Review*, 2022, *112* (1), 213–234.

Kleven, Henrik Jacobsen, Claus Thustrup Kreiner, and Emmanuel Saez, "Why can modern governments tax so much? An agency model of firms as fiscal intermediaries," *Economica*, 2016, *83* (330), 219–246.

_ , Martin B Knudsen, Claus Thustrup Kreiner, Søren Pedersen, and Emmanuel Saez, "Unwilling or unable to cheat? Evidence from a tax audit experiment in Denmark," *Econometrica*, 2011, *79* (3), 651–692.

Kumler, Todd, Eric Verhoogen, and Judith Frías, "Enlisting employees in improving payroll tax compliance: Evidence from mexico," *Review of Economics and Statistics*, 2020, *102* (5), 881–896.

Lok Sabha Questions, "Unstarred Question No. 435: Unearthing of Fake Input Tax Credit," Answered by the Minister of State in the Ministry of Finance, https://sansad.in/getFile/loksabhaquestions/annex/1715/AU435.pdf?source=pqals 2024.

Mascagni, Giulia, Roel Dom, Fabrizio Santoro, and Denis Mukama, "The VAT in practice: equity, enforcement, and complexity," *International Tax and Public Finance*, 2023, *30* (2), 525–563.

Meiselman, Ben S, "Ghostbusting in Detroit: Evidence on nonfilers from a controlled field experiment," *Journal of Public Economics*, 2018, *158*, 180–193.

Mittal, Shekhar, Ofir Reich, and Aprajit Mahajan, "Who is bogus? using one-sided labels to identify fraudulent firms from tax returns," in "Proceedings of the 1st ACM SIGCAS Conference on Computing and Sustainable Societies" 2018, pp. 1–11.

Okunogbe, Oyebola and Gabriel Tourek, "How can lower-income countries collect more taxes? The role of technology, tax agents, and politics," *Journal of Economic Perspectives*, 2024, *38* (1), 81–106.

_ and Victor Pouliquen, "Technology, taxation, and corruption: evidence from the introduction of electronic tax filing," *American Economic Journal: Economic Policy*, 2022, *14* (1), 341–372.

Pomeranz, Dina, "No taxation without information: Deterrence and self-enforcement in the value added tax," *American Economic Review*, 2015, *105* (8), 2539–2569.

Rubin, Donald B, "The Bayesian Bootstrap," *The Annals of Statistics*, 1981, pp. 130–134.

Shah, Jawad, "Using computerized information to enforce VAT: Evidence from Pakistan," *Journal of Development Economics*, 2026, *178*, 103595.

Waseem, Mazhar, "Overclaimed refunds, undeclared sales, and invoice mills: Nature and extent of noncompliance in a value-added tax," *Journal of Public Economics*, 2023, *218*, 104783.

Table 1: Effect on Filing Behavior of Fake Credit Generators

	(1)	(2)	(3)
	Filed Credit-Linked Return	Filed Payment-Linked Return	Filed Neither Return
Panel A: Non-Filer Fake Credit Generators			
Fake Credit Generator × Post	-0.109	-0.002	0.097
	(0.017)	(0.014)	(0.017)
Outcome Mean	0.811	0.627	0.164
Observations	57,372	57,372	57,372
Panel B: Filer Fake Credit Generators			
Fake Credit Generator × Post	-0.020	-0.007	0.006
	(0.002)	(0.002)	(0.001)
Outcome Mean	0.959	0.999	0.000
Observations	2,806,552	2,806,552	2,806,552
Panel C: All Fake Credit Generators			
Fake Credit Generator × Post	-0.029	-0.007	0.015
	(0.003)	(0.002)	(0.002)
Outcome Mean	0.945	0.964	0.016
Observations	2,858,030	2,858,030	2,858,030

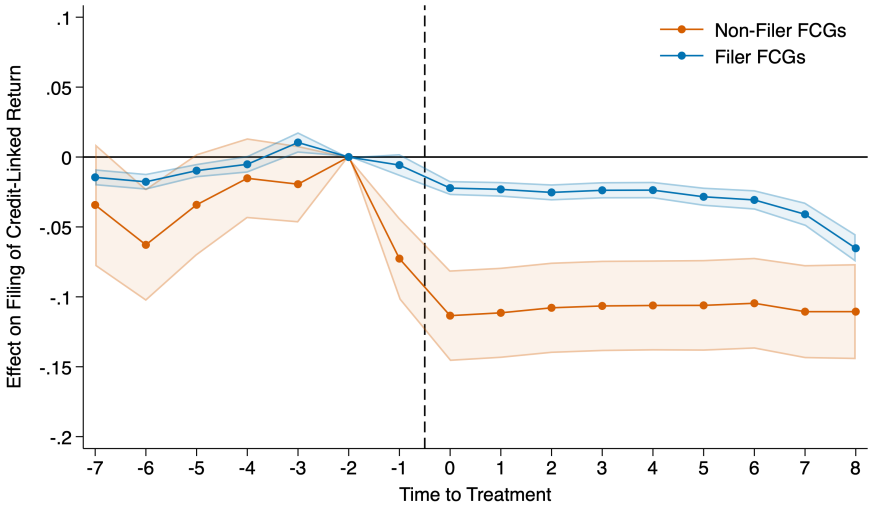
Notes: This table presents the causal effect of the policy on filing outcomes of Fake Credit Generators (FCGs). Panel A compares Non-Filer FCGs, i.e., FCGs that do not file at least one payment-linked return between January-June 2021, to a matched control group of non-FCGs that also do not file at least one payment-linked return between January-June 2021. Panel B compares Filer FCGs, i.e., FCGs that file all payment-linked returns between January-June 2021, to a matched control group of non-FCGs that also file all payment-linked returns between January-June 2021. Panel C compares all FCGs to a matched control group of all non-FCGs. In each panel, the control group is constructed by matching on pre-treatment firm characteristics (age, size, nature of business, whether the firm reports inter-state sales, and the number of filed pre-period payment returns) using Coarsened Exact Matching (CEM). The matching in Panels A and B is performed separately within the non-filer and filer subsamples, respectively. Column 1 presents the effect on filing of the credit-linked return in a given month. Column 2 presents the effect on filing of the payment-linked return in a given month. Column 3 presents the effect on filing neither return in a given month (where the outcome is equal to 1 if a firm files neither return, and 0 otherwise). In each case, the causal effect is estimated using the two-way fixed effects approach we describe in Section 4. We exclude all data after February, 2022 to remove any effect of delayed filing on the estimated treatment effect. The outcome mean is the average of the outcome for the treatment group before the policy was implemented. Standard errors are clustered at the firm level.

Table 2: Effect on Network Firms

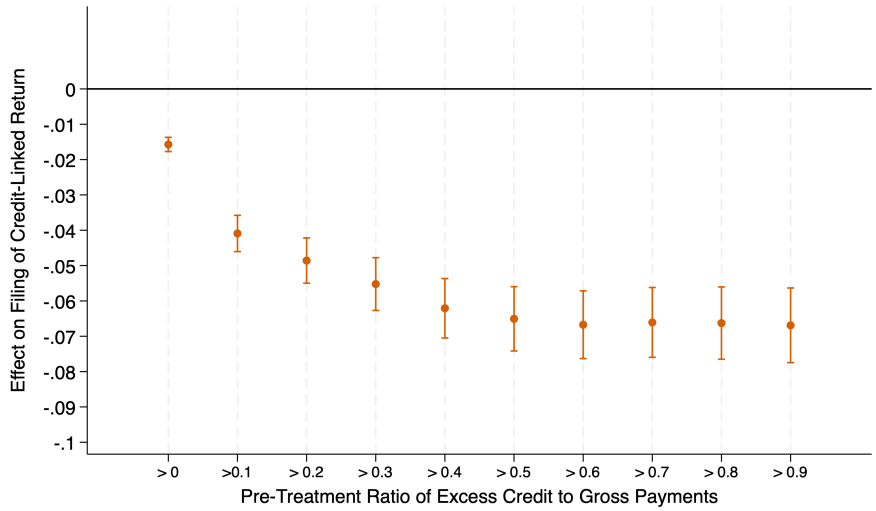
	(1)	(2)	(3)
	Net Tax Payments (Share)	Credit Used (Share)	Log Gross Payments
Panel A: Network of Non-Filer Fake Credit Generators			
Network Firm × Post	0.017	-0.017	-0.022
	(0.005)	(0.005)	(0.028)
Outcome Mean	0.181	0.819	11.799
Observations	1,875,425	1,875,425	1,875,425
Panel B: Network of Filer Fake Credit Generators			
Network Firm × Post	0.013	-0.013	-0.017
	(0.002)	(0.002)	(0.013)
Outcome Mean	0.272	0.728	11.870
Observations	1,944,395	1,944,395	1,944,395
Panel C: Network of All Fake Credit Generators			
Network Firm × Post	0.014	-0.014	-0.017
	(0.002)	(0.002)	(0.012)
Outcome Mean	0.259	0.741	11.862
Observations	1,957,302	1,957,302	1,957,302

Notes: This table presents the causal effect of the policy on the network (buyers) of firms we classify as Fake Credit Generators (FCGs). We define the network of a group of FCGs as those firms that receive at least 5% of their total input tax credit between January-June 2021 from FCGs in that group. Firms that receive a positive but smaller share of their input tax credit from these FCGs are excluded from the analysis. In each panel, we compare the network firms to a control group of all firms that receive none of their input tax credit between January-June 2021 from any FCG. We exclude all FCGs themselves and the clients of FCGs that file their returns quarterly from the analysis. Panel A focuses on the network of Non-Filer FCGs, i.e., FCGs that do not file at least one payment-linked return between January-June 2021. Firms that receive input tax credit from FCGs but not from Non-Filer FCGs are excluded from this panel. Panel B focuses on the network of Filer FCGs, i.e., FCGs that file all payment-linked returns between January-June 2021. Firms that receive input tax credit from FCGs but not from Filer FCGs are excluded from this panel. Panel C focuses on the network of all FCGs. The outcome in Column 1 is defined as the ratio of net tax payments made by a firm using cash divided by the firm's gross tax payments. The outcome in Column 2 is defined as the ratio of tax payments made by a firm using input tax credit divided by the firm's gross tax payments. The outcome in Column 3 is defined as the log of the firm's gross tax payments in a given time period. A firm reporting zero gross tax payments in any month will have missing values for all the outcomes in that month. We estimate the causal effects using the two-way fixed effects design described in Section 4. The outcome mean is the average of the outcome for the network firms in each panel before the policy was implemented. Standard errors are clustered at the firm-level.

Figure 1: Effect of Policy on Filing of Credit-Linked Return



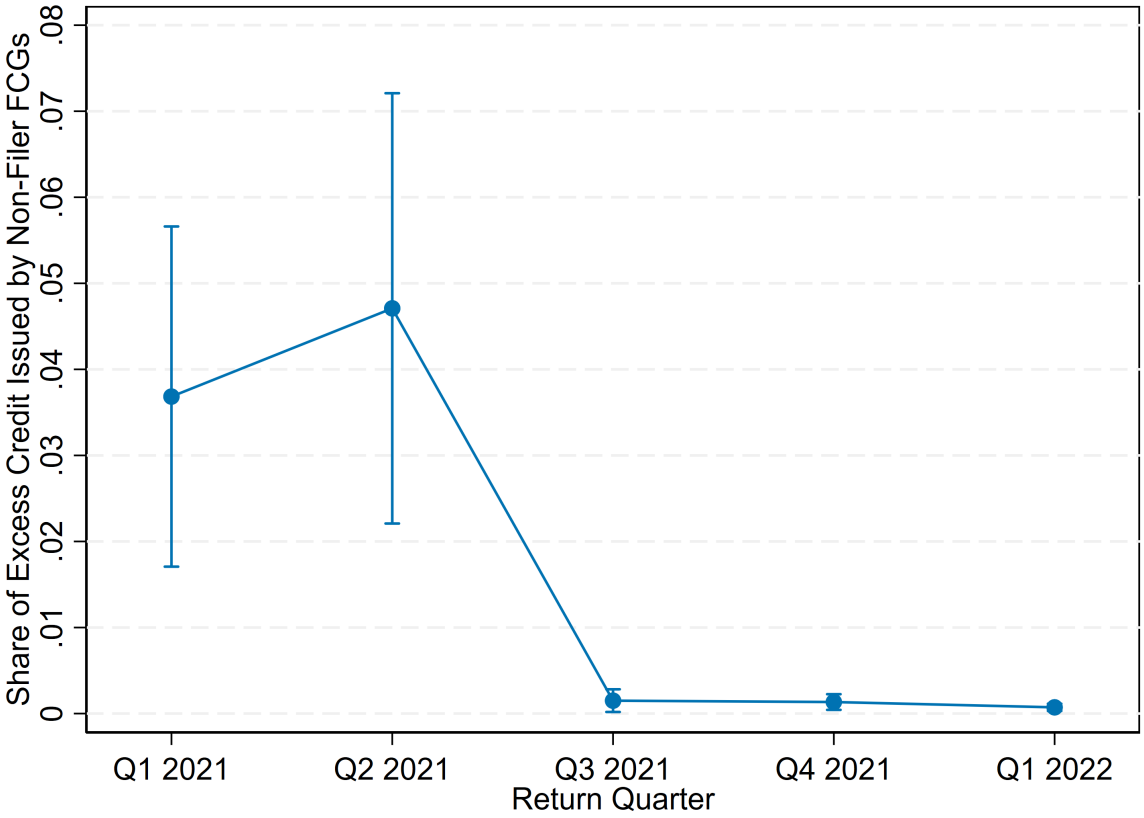
(a) Event Study



(b) Heterogeneity by Pre-Period Discrepancy

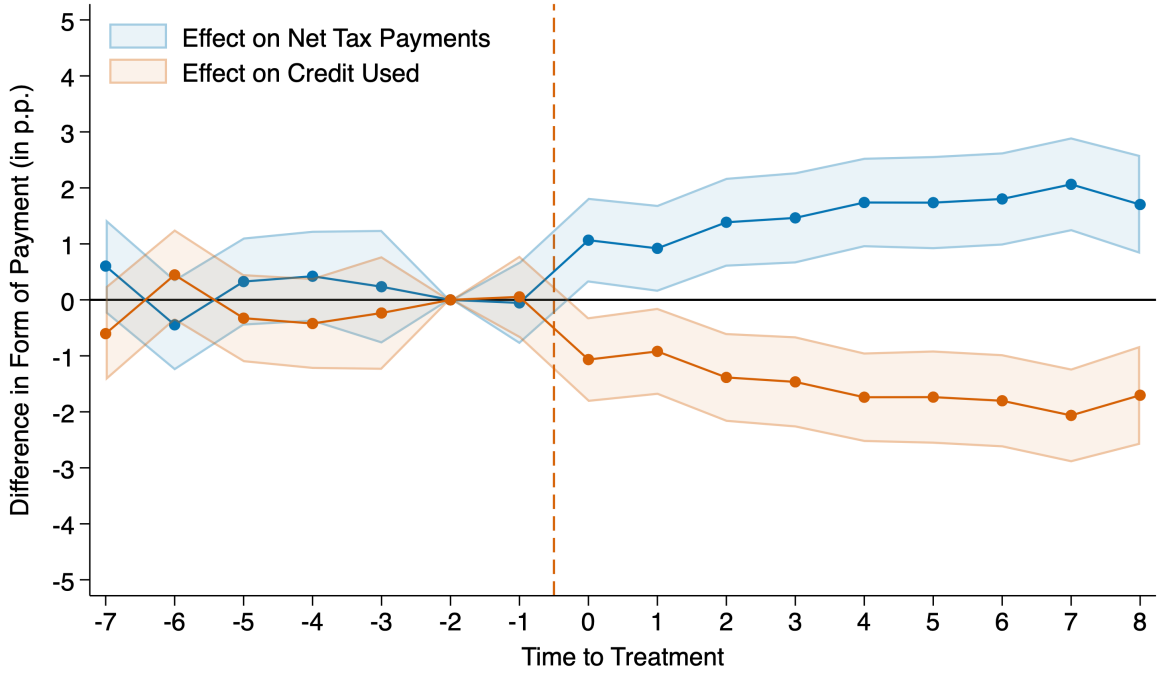
Notes. This figure presents the causal effect of the policy on the filing of the credit-linked return by Fake Credit Generators (FCGs). Panel A plots the event-study coefficients estimated from Equation 1 in Section 4, separately for Non-Filer FCGs and Filer FCGs. Non-Filer FCGs are the firms that we classify as FCGs who additionally do not file at least one payment-linked return between January-June 2021; their control group comprises all firms not classified as FCGs that also do not file at least one payment-linked return between January-June 2021. Filer FCGs are the firms that we classify as FCGs who file all payment-linked returns between January-June 2021; their control group comprises all firms not classified as FCGs that also file all payment-linked returns between January-June 2021. Panel B plots the estimated Average Treatment Effect on the Treated (ATT) of the policy on the filing of the credit-linked returns, estimated separately across different pre-treatment ratios of Total Excess Credit to Total Gross Payments. Total Gross Payments are defined as the sum of all tax payments made by the firm between January and June 2021. Total Excess Credit is defined as the total credit issued by a firm to its buyers between January and June 2021, minus the Total Gross Payments made by the firm over the same period. For the first coefficient plotted in Panel B, the treatment group consists of FCGs and firms with positive excess credit that was small enough to be excluded from the FCG classification. For each subsequent coefficient, the treatment group is restricted to FCGs with a ratio of Total Excess Credit to Total Gross Payments greater than the threshold value displayed on the x-axis, while the control group is fixed to include all firms with zero excess credits during the January to June 2021 period. The sample in Panel B excludes all firms with no gross payments, and additionally excludes all data after February 2022 to eliminate any confounding effects of delayed filing on the estimated treatment effects. In both panels, standard errors are clustered at the firm level, and 95 percent confidence intervals are displayed.

Figure 2: Trends in Excess Credit Attributable to Non-Filer Fake Credit Generators



Notes. This figure plots the proportion of Excess Credit in a given quarter that is attributable to Non-Filer Fake Credit Generators (FCGs) in our primary sample, described in Section 3. Excess Credit for a firm is defined as the total credit issued by the firm to their clients in that quarter minus its total tax payments in that quarter. Non-Filer FCGs are the firms that we classify as Fake Credit Generators who additionally don't file at least one payment-linked return between January-June 2021. In this figure, we show the sum of Excess Credits attributable to Non-Filer Fake Credit Generators in a quarter divided by the total amount of Excess Credit in the economy in that quarter

Figure 3: Effect of Policy on Tax Payments by Network Firms



Notes. This figure shows the causal effect of the policy on the tax payment method used by the network of firms we classify as Fake Credit Generators. We define the network of Fake Credit Generators as those firms who receive at least 5% of their total input tax credits between January-June 2021 from Fake Credit Generators. We compare these network firms to all firms who receive none of their input tax credits between January-June 2021 from Fake Credit Generators. We exclude all Fake Credit Generators themselves from this analysis. The outcome is defined as the ratio of tax payments made using cash or input tax credits to the total tax payments in that return period. For a given firm, this ratio must necessarily sum to one in a given return period. We plot the event-study coefficients from Equation 1 described in Section 4 along with the 95% confidence intervals, defining the network of FCGs as the treated firms. Standard errors are clustered at the firm-level.

Appendix

A Appendix Figures and Tables

Table A.1: Summary Statistics

	(1)	(2)	(3)	(4)	(5)
	Control Firms	Fake Credit Generators	Non-Filer Fake Credit Generators	Difference: (2)-(1)	P-Value: (2)-(1)
Number of Credit-Linked Returns Filed	5.718	5.713	5.008	-0.006	0.598
Number of Payment-Linked Returns Filed	5.913	5.765	3.720	-0.148	0.000
Filed All Pre-Period Payment-Linked Returns	0.974	0.897	0.000	-0.077	0.000
Age (in months)	30.629	31.335	26.247	0.705	0.000
Has Inter-State Sales	0.653	0.692	0.623	0.039	0.000
Credit Issued	3,027,093	4,671,994	2,521,705	1,644,901	0.062
Gross Payments	4,115,650	3,261,315	1,514,462	-854,335	0.253
Net Payments	830,632	774,941	131,415	-55,692	0.726
Nature of Business					
Retail	0.328	0.217	0.273	-0.110	0.000
Services	0.176	0.316	0.193	0.139	0.000
Wholesale/Manufacturing	0.240	0.182	0.283	-0.058	0.000
Other	0.256	0.285	0.250	0.029	0.000
Number of Firms	200,868	5,811	600		

Notes: This table reports the summary statistics for all firms in our baseline sample, using data before the policy was implemented between January-June 2021. Column 1 reports the mean value of the corresponding variable for the firms we classify to be in the control group, i.e. firms that are not fake credit generators. Column 2 reports the mean value of the corresponding variable for the firms we classify to be Fake Credit Generators. Column 3 reports the mean value of the corresponding variable for the subset of Fake Credit Generators that additionally do not file at least one payment-linked return between January-June 2021, i.e., the Non-Filer FCGs. Column 4 reports the difference between Column 1 and Column 2. Column 5 provides the p-value from a t-test of the difference between Column 1 and Column 2. "Filed All Pre-Period Payment-Linked Returns" is equal to 1 if a firm filed all payment-linked returns between January-June 2021. "Age" refers to the number of months the firm had been registered for at the start of our sample period (Jan 2021). Because the GST system was introduced in July 2017, the largest value the age variable can take is 42. "Has Inter-State Sales" takes a value of 1 if a firm has any buyer that is not registered in New Delhi, and 0 otherwise. "Credit Issued" is the total rupee value of input tax credits supplied by the firm to their buyers between Jan-June 2021. "Gross Payments" is the total rupee value of tax payments made by the firm between Jan-June 2021. "Net Payments" is the total rupee value of tax payments made by the firm in cash between Jan-June 2021.

Table A.2: Effect on Being Inactive for Past 3 Months

	(1)
	Inactive For 3 Months
Fake Credit Generator × Post	0.014
	(0.003)
Outcome Mean	0.008
Observations	2,041,450

Notes: This table presents the causal effect of the policy on filing neither form for the past 3 months (where the outcome is equal to 1 at time t if a firm files neither form at t,$t-1$, and $t-2$, and 0 otherwise). To obtain clean pre- and post-policy periods, we exclude the return periods August and September 2021 from this analysis. The causal effect is estimated using the two-way fixed effects approach we describe in Section 4. We use a matched control group, where we match on pre-treatment firm characteristics (age, size, nature of business, whether the firm reports inter-state sales, and the number of filed pre-period payment returns) using Coarsened Exact Matching (CEM). We exclude all data after February, 2022 to remove any effect of delayed filing on the estimated treatment effect. Standard errors are clustered at the firm level. The outcome mean is the average of the outcome for the treatment group before the policy was implemented.

Table A.3: Effect on Filing of Credit-Linked Return in Placebo Treatment Group

	(1)
	Filed Credit-Linked Return
Placebo FCG × Post	-0.004
	(0.001)
Outcome Mean	0.975
Observations	2,493,134

Notes: This table presents the causal effect of the policy on filing of a credit-linked return by firms with a positive but negligible discrepancy between credit issued and gross payments. Specifically, the treatment status is set to 0 for all firms who did not issue credit in excess of gross payments between Jan-June 2021, and it is set to 1 for all firms who issued between 0-10% more credit than their gross payments between Jan-June 2021. All firms with a discrepancy larger than 10% are dropped from this analysis. The outcome is equal to 1 in a given return period if a firm filed the credit-linked return and is 0 otherwise. The causal effect is estimated using the two-way fixed effects approach we describe in Section 4. We exclude all data after February, 2022 to remove any effect of delayed filing on the estimated treatment effect. Standard errors are clustered at the firm level. The outcome mean is the average of the outcome for the treatment group before the policy was implemented.

Table A.4: Summary Statistics - Network Firms

	(1)	(2)	(3)	(4)	(5)
	Control Firms	Network Firms	Non-Filer Network Firms	Difference: (2)-(1)	P-Value: (2)-(1)
Number of Credit-Linked Returns Filed	5.705	5.785	5.774	0.080	0.000
Number of Payment-Linked Returns Filed	5.899	5.980	5.975	0.081	0.000
Age (in months)	29.514	34.715	34.813	5.201	0.000
Has Inter-State Sales	0.608	0.825	0.876	0.216	0.000
Credit Issued	1,478,672	11,064,002	4,249,417	9,585,330	0.000
Gross Payments	1,878,889	17,727,650	5,259,054	15,848,761	0.000
Net Payments	312,964	5,185,924	473,260	4,872,959	0.000
Ratio of Net Tax Payments to Gross Payments	0.320	0.259	0.182	-0.060	0.000
Nature of Business					
Retail	0.328	0.308	0.371	-0.020	0.000
Services	0.184	0.192	0.099	0.008	0.096
Wholesale/Manufacturing	0.231	0.238	0.326	0.008	0.147
Other	0.257	0.262	0.203	0.005	0.361
Number of Firms	170,389	6,973	929		

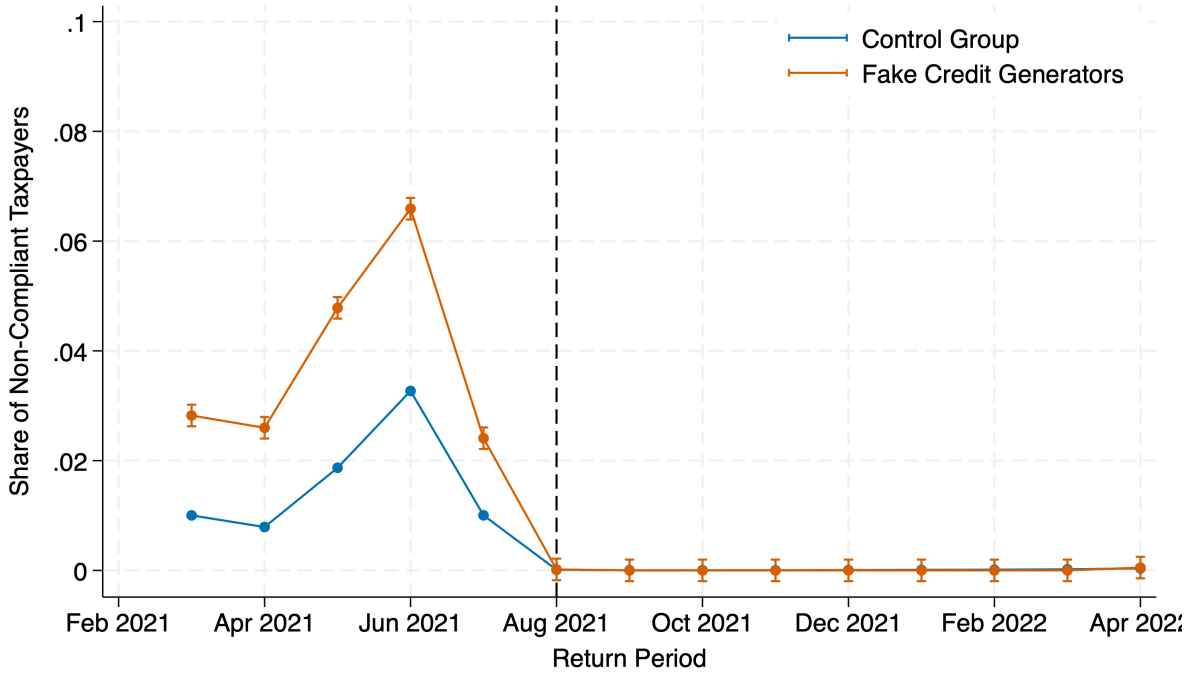
Notes: This table reports the summary statistics for the network of fake credit generators, using data before the treatment was implemented between January-June 2021. Column 1 reports the mean value of the corresponding variable for the firms that don't receive any input tax credit from fake credit generators between January-June 2021. Column 2 reports the mean value of the corresponding variable for the firms that receive at least 5% of their input tax credit from Fake Credit Generators. Column 3 reports the mean value of the corresponding variable for the firms that receive at least 5% of their input tax credit from Fake Credit Generators that additionally do not file at least one payment-linked return between January-June 2021. Column 4 reports the difference between Column 1 and Column 2. Column 5 provides the p-value from a t-test of the difference between Column 1 and Column 2. "Age" refers to the number of months the firm had been registered for at the start of our sample period (Jan 2021). Because the GST system was introduced in July 2017, the largest value the age variable can take is 42. "Has Inter-State Sales" takes a value of 1 if a firm has any buyer that is not registered in New Delhi, and 0 otherwise. "Credit Issued" is the total rupee value of input tax credits supplied by the firm to their buyers between Jan-June 2021. "Gross Payments" is the total rupee value of tax payments made by the firm between Jan-June 2021. "Net Payments" is the total rupee value of tax payments made by the firm in cash between Jan-June 2021.

Table A.5: Effect on Any Sales of Network Firms

	(1)
	Any Sales
Network Firm × Post	-0.004
	(0.002)
Outcome Mean	0.859
Observations	2,837,792

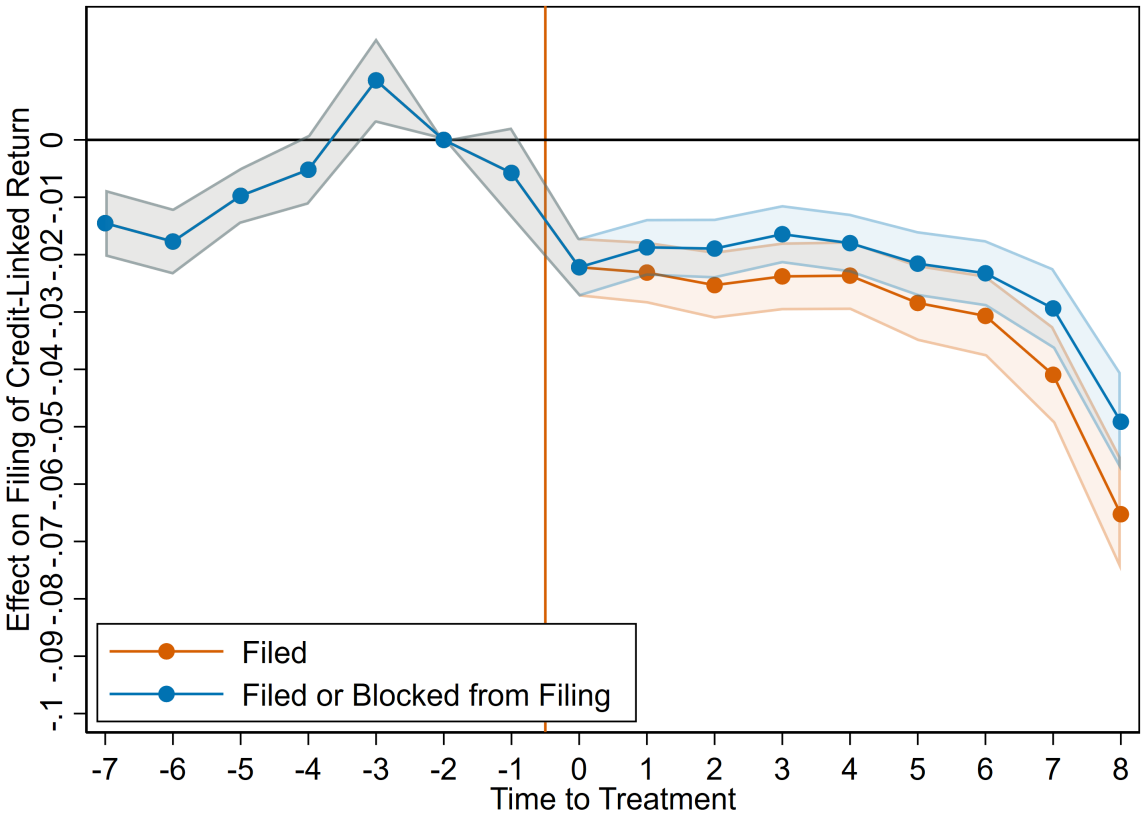
Notes: This table presents the causal effect of the policy on the network (buyers) of firms we classify as Fake Credit Generators. We define the network of Fake Credit Generators as those firms who receive at least 5% of their total input tax credit between January-June 2021 from Fake Credit Generators. We compare these network firms to all firms who receive none of their input tax credits between January-June 2021 from Fake Credit Generators. We exclude all Fake Credit Generators themselves from this analysis. The outcome is equal to 1 if the network firm reported non-zero sales in that return period, and 0 otherwise. Standard errors are clustered at the firm level. The outcome mean is the average of the outcome for the treatment group before the policy was implemented.

Figure A.1: Pre-Policy Exposure to Blocking Rule



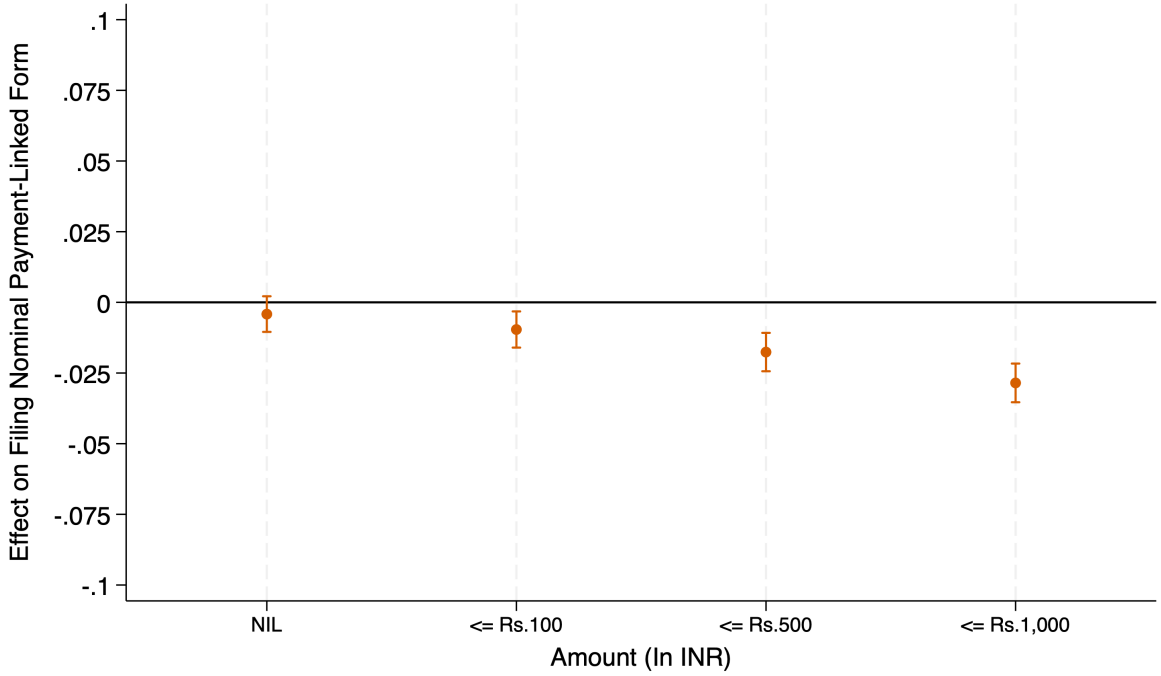
Notes. This figure plots the proportion of taxpayers that are non-compliant with the policy described in Section 2. Specifically, this figure shows the proportion of taxpayers who satisfy the following two conditions: (a) file a credit-linked return in a given return period, and (b) had not filed a payment-linked return for the preceding two periods at the time of filing the credit-linked return.

Figure A.2: Filer FCGs: Behavioral Response vs. Mechanical Blocking



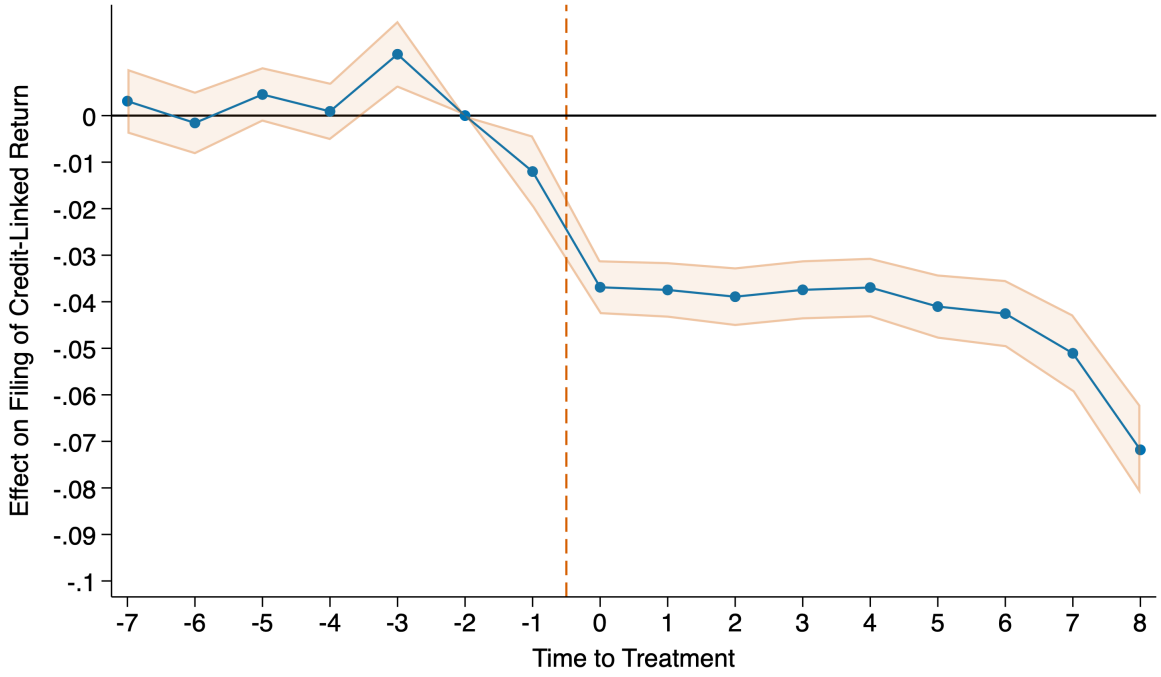
Notes. This figure shows the causal effect on filing of the credit-linked return by Filer Fake Credit Generators. Filer Fake Credit Generators are the firms that we classify as Fake Credit Generators who additionally file all payment-linked returns between January-June 2021. We plot the event-study coefficients from Equation 1 described in Section 4, along with the 95% confidence intervals, for two outcomes. The first outcome ("Filed") is equal to 1 if the firm filed the credit-linked return for that return period, and 0 otherwise. The second outcome ("Filed or Blocked from Filing") is equal to 1 if the firm either filed the credit-linked return for that return period or was blocked from filing it, where a firm is blocked if it had not filed a payment-linked return for the preceding two periods at the time of filing the credit-linked return. The control group includes all firms we do not classify as Fake Credit Generators that additionally file all payment-linked returns between January-June 2021. Filing outcomes in the periods 7 and 8 are affected by delayed filing and truncation of our data extracted in June 2022. Standard errors are clustered at the firm-level.

Figure A.3: Effect on Filing of Token Payment-Linked Returns



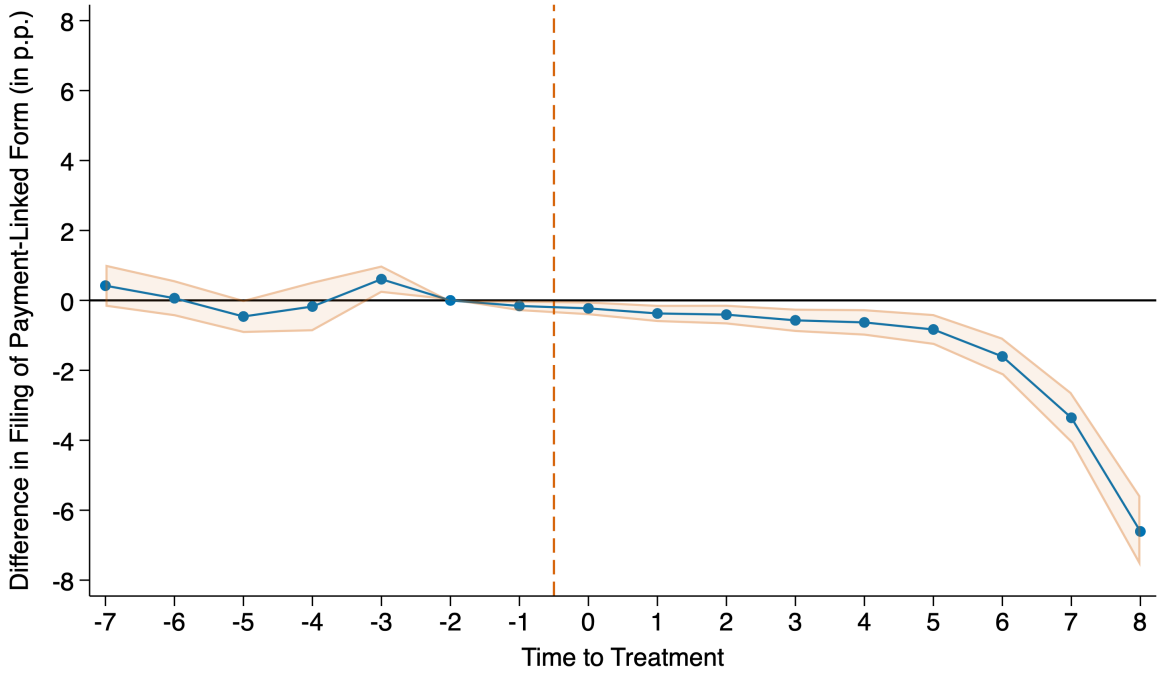
Notes. This figure presents the causal effect of the policy on filing of a token payment-linked return among all FCGs compared to the control group. Specifically, the outcome is equal to 1 in a given return period if a firm filed the payment-linked return and made gross payments lower than or equal to the amount on the x-axis, and is 0 otherwise. A NIL amount corresponds to no gross payments made with a filed payment-linked return. The causal effect is estimated using the two-way fixed effects approach we describe in Section 4. We use a matched control group, where we match on pre-treatment firm characteristics (age, size, nature of business, whether the firm reports inter-state sales, and the number of filed pre-period payment returns) using Coarsened Exact Matching (CEM). We exclude all data after February 2022 to remove any effect of delayed filing on the estimated treatment effect. Standard errors are clustered at the firm level.

Figure A.4: Effect on Filing of Credit-Linked Return by All Fake Credit Generators



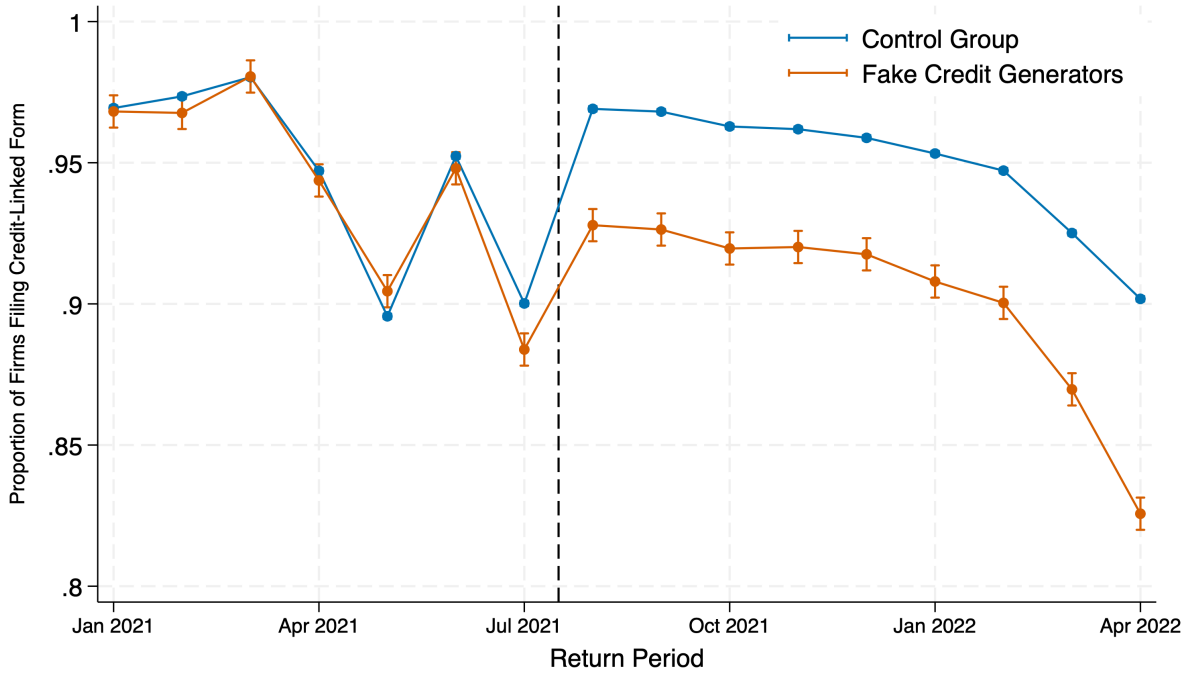
Notes. This figure shows the causal effect of the policy on filing of the credit-linked return by all firms we classify as Fake Credit Generators. We plot the event-study coefficients from Equation 1 described in Section 4, along with the 95% confidence intervals. The control group includes all firms we do not classify as Fake Credit Generators. Filing outcomes in the periods 7 and 8 are affected by delayed filing and truncation of our data extracted in June 2022. Standard errors are clustered at the firm-level.

Figure A.5: Effect of Policy on Filing of Payment-Linked Return



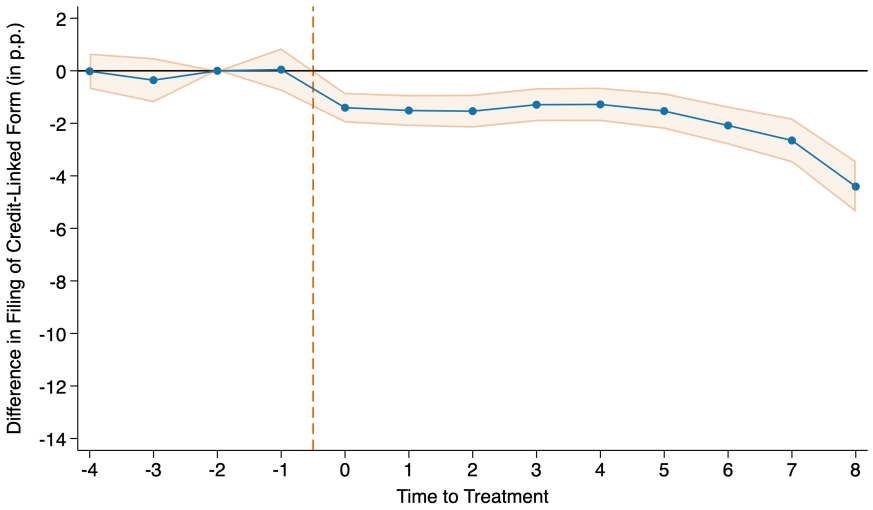
Notes. This figure shows the causal effect of the policy on filing of the payment-linked return. We plot the event-study coefficients from Equation 1 described in Section 4, along with the 95% confidence intervals. The treatment group consists of all firms we classify as Fake Credit Generators. We use a matched control group, where we match on pre-treatment firm characteristics (age, size, nature of business, whether the firm reports inter-state sales, and the number of filed pre-period payment returns) using Coarsened Exact Matching (CEM). Filing outcomes in the periods 7 and 8 are affected by delayed filing and truncation of our data extracted in June 2022. Standard errors are clustered at the firm-level.

Figure A.6: Trends in Filing of Credit-Linked Return

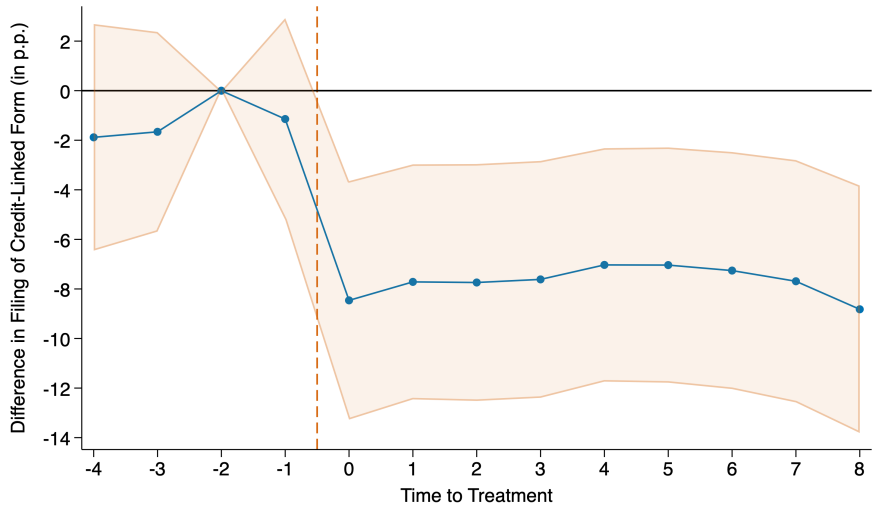


Notes. This figure plots the proportion of firms that filed the credit-linked return in a given return period along with the 95% confidence intervals, separately for firms we classify as Fake Credit Generators and firms we do not classify as Fake Credit Generators (the control group). Filing outcomes in the periods 7 and 8 are affected by delayed filing and truncation of our data extracted in June 2022.

Figure A.7: Effect of Policy on Filing of Credit-Linked Return (Testing for Mean Reversion)



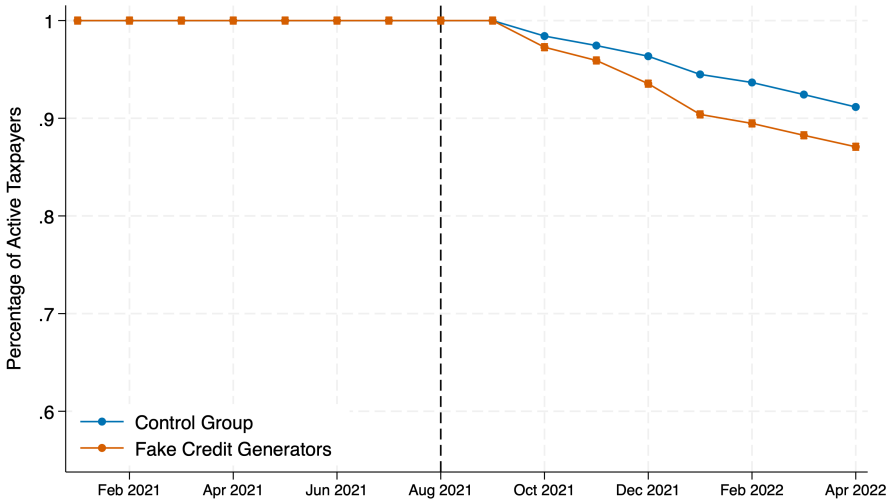
(a) Overall Sample



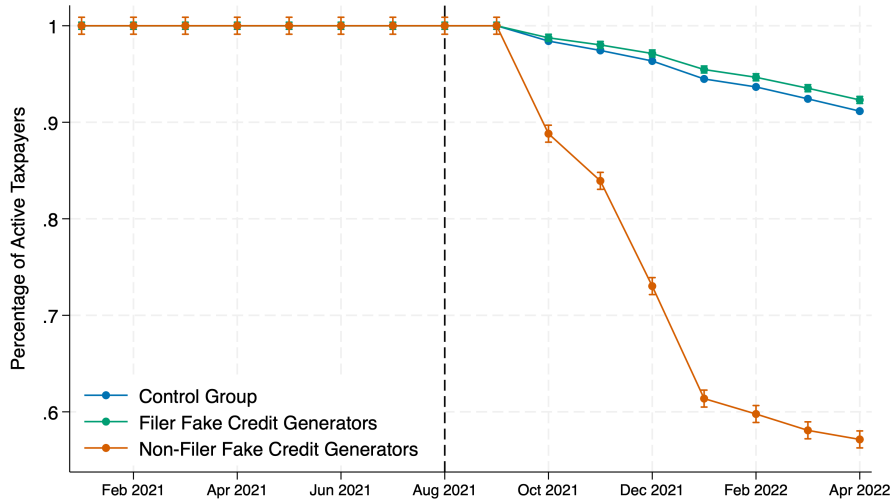
(b) Pre-Period Non-Filers

Notes. This figure shows the causal effect of the policy on filing of the credit-linked return by Fake Credit Generators. This figure classifies firms as Fake Credit Generators only using data from January-March 2021, using the same criteria described in Section 3.1. The causal effect is estimated with an event-study design using data from April 2021-April 2022. We plot the event-study coefficients along with the 95% confidence intervals. Panel A compares all firms classified as Fake Credit Generators using data from January-March 2021 to all firms not classified as Fake Credit Generators. Panel B restricts the analysis to firms that additionally don't file at least one payment-linked return between January-June 2021. Filing outcomes in the periods 7 and 8 are affected by delayed filing and truncation of our data extracted in June 2022. Standard errors are clustered at the firm-level.

Figure A.8: Trends in Registration Status of Firms



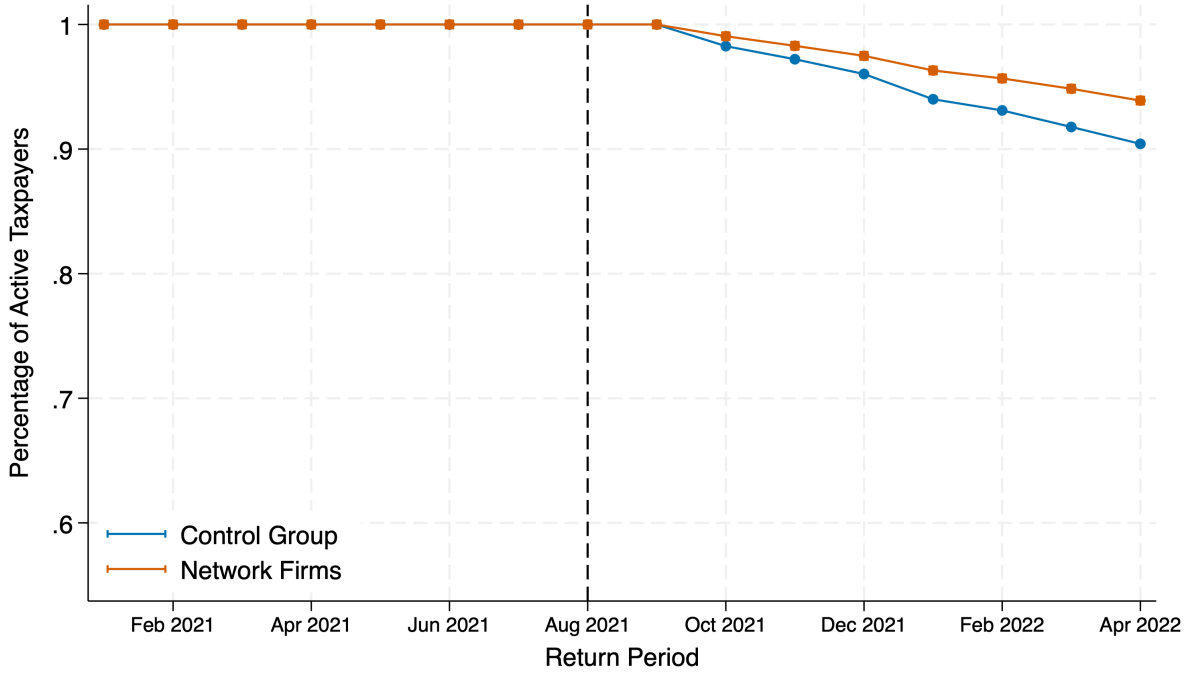
(a) Share of Active Firms



(b) Share of Active Firms by Fake Credit Generator Type

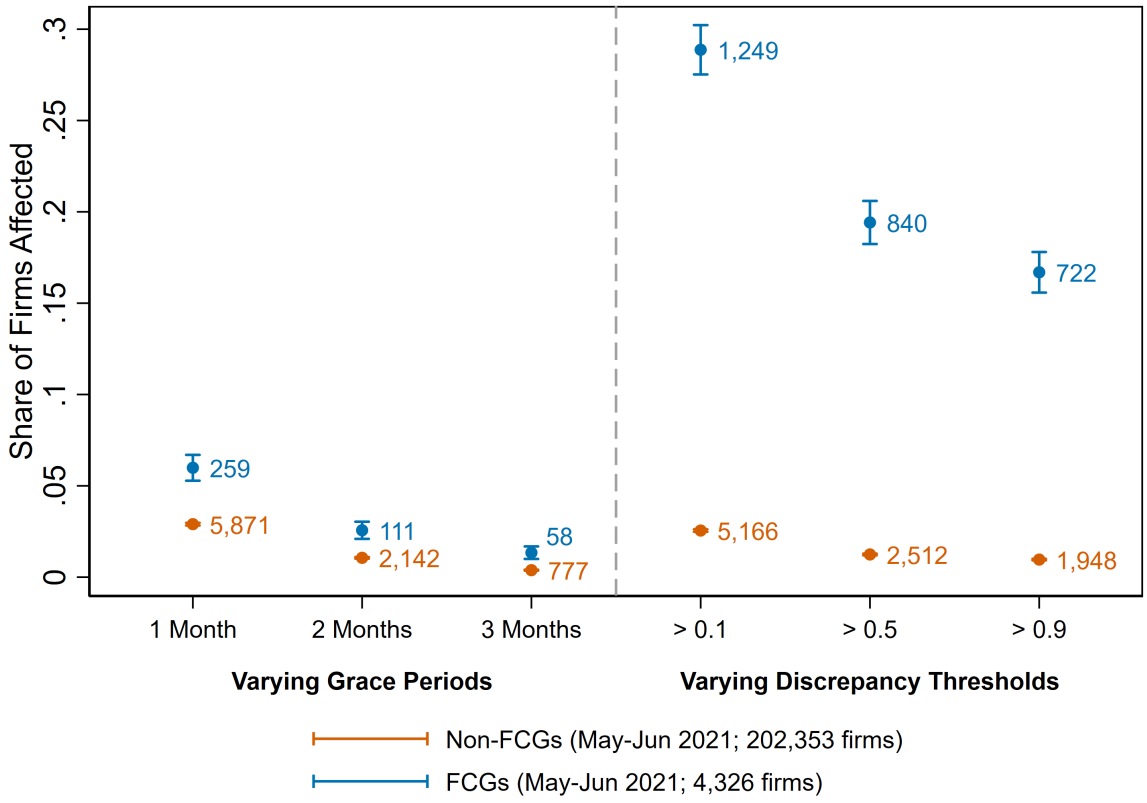
Notes. This figure presents trends in registration status of firms in our expanded sample as described in Section 3 with an additional restriction that they are registered through September 2021. This is to avoid imputing month of deregistration in missing months because of the registration data restriction described in Section 3. We classify firms as Fake Credit Generators using the procedure described in Section 3.1. In Panel A, we present the proportion of Fake Credit Generator firms (compared to a control group of all other firms) that are registered in each return period. In Panel B, we separately show the proportion of registered firms for Non-Filer FCGs (that don't file at least one payment-linked return between January-June 2021) and Filer FCGs (that file all payment-linked returns between January-June 2021).

Figure A.9: Trends in Registration Status of Network Firms



Notes. This figure presents trends in registration status of firms in our expanded sample as described in Section 3 with an additional restriction that they are registered through September 2021. This is to avoid imputing month of deregistration in missing months because of the registration data restriction described in Section 3. We classify firms as Fake Credit Generators using the procedure described in Section 3.1. We define the network of these Fake Credit Generators as those firms who receive at least 5% of their total input tax credit between January-June 2021 from Fake Credit Generators and are registered in all periods between January-September 2021. We compare these network firms to all firms who receive none of their input tax credits between January-June 2021 from Fake Credit Generators and are registered in all periods between January-September 2021.

Figure A.10: Share of Firms Affected By Counterfactual Policy Triggers



Notes. This figure displays the share of firms that would be affected by two types of policy triggers, reported separately for Fake Credit Generators (FCGs) and non-FCGs. Firms are classified as FCGs or non-FCGs based on their behavior in May–June 2021, following the procedure detailed in Section 3.1, so that the classification window does not overlap with the January–April 2021 data used to determine which firms would be affected. The resulting number of FCGs and non-FCGs is reported in parentheses in the legend. Each point is the share of that group affected by the corresponding policy, and the number printed beside each point is the count of affected firms. The left panel ("Varying Grace Periods") considers blocking rules that condition the filing of the credit-linked return on the filing of prior payment-linked returns. Using data from January through April 2021, we identify firms that (a) filed their February credit-linked return while their January payment-linked return was pending, (b) filed their March credit-linked return while the payment-linked returns for the preceding two months were pending, and (c) filed their April credit-linked return while the payment-linked returns for the preceding three months were pending. A payment-linked return is pending if it had not been filed, or was filed only after the credit-linked return in question. A policy allowing a grace period of less than one month between the filing of the payment- and credit-linked returns would affect firms in groups (a), (b), and (c) ("1 Month"); a grace period of less than two months would affect firms in groups (b) and (c) ("2 Months"); and a grace period of less than three months would affect only firms in group (c) ("3 Months"). To ensure comparability across counterfactual grace periods, we only count firms whose non-compliance begins with the January 2021 payment-linked return. Each rule is then evaluated on the same set of firms and the rules differ only in how many consecutive payment-linked returns must be pending before they bind. The right panel ("Varying Discrepancy Thresholds") considers rules that trigger when a firm's ratio of Total Excess Credit to Total Gross Payments over January–April 2021 exceeds a threshold (>0.1, >0.5, or >0.9). Total Gross Payments is the sum of all tax payments made by the firm during January–April 2021; Total Excess Credit is the total credit the firm issued to its buyers during January–April 2021 minus its Total Gross Payments over the same period. Firms with no gross payments over this period are counted as affected at every threshold if they issued any excess credit and as unaffected otherwise. The 95% confidence intervals for each share are presented.

B Alternative Samples

Our main analysis focuses on a balanced sample of monthly filers who are registered throughout the study period of January 2021 - April 2022. In this section, we investigate the effect of the policy on two alternative samples, accounting for firms that deregistered within this study period and firms that elect to file quarterly—rather than monthly—returns.

First, we broaden our definition of FCGs to include firms that were registered for at least one return period between January and June 2021, so that they can be characterized as FCGs based on their discrepancies between payment-linked and credit-linked returns in the pre-policy window. Once registered, all firms remain in the sample in each following return period regardless of their contemporaneous registration status. Our outcomes of interest, whether the firm filed a payment-linked or credit-linked return, take a value of zero if the taxpayer did not file the return whether that be because they were no longer registered or if they were registered but non-compliant. Expanding the sample in this manner increases the magnitude of the estimated response to the policy. Appendix Table B.1 shows the main results for this expanded sample. Column 1 shows that the FCGs increase their non-filing rate of credit-linked return by 6.2 percentage points. Figure B.1 shows the event study. We also find that similar to the main analysis, there is a slight decrease in filing of payment-linked return (Column 2). When firms stop filing their returns as a result of the policy, they are more likely to become deregistered and drop out of our main sample: we observe a higher probability of filing neither return among FCGs after the blocking policy.

In column 4 we estimate an increase of 1.9 percentage points in the likelihood of deregistration itself, consistent with the rise in inactivity. One caveat about this estimate is that we cannot verify parallel pre-trends for this outcome. Registration data was provided in monthly snapshots except in the months between February-June 2021 and August 2021. We do not know the exact month of deregistration for taxpayers deregistered in these months and so do not have sufficient monthly pre-period deregistration rates to estimate an event-study.

We trace the downstream effects of the disruption of fraudulent credit on firms that are its beneficiaries, as in Section 5.2. Table B.2 shows these results, and Figure B.2 shows the event study. We find similar effects, as in the main sample. We do observe a small negative effect on gross tax payments (Column 3) and account for this potential decline in tax payments when computing the aggregate revenue implications of the policy in Section 5.3.

Next, we examine outcomes among quarterly filers who were excluded from the main analysis sample because we observe them for only two quarters before the policy. Only firms with annual revenue below INR 50 million can file on a quarterly basis, making this group of firms quite different from the main sample. They may be more likely to make legitimate mistakes in filing both their payment- and credit-linked returns, as they must reconcile information over three months instead of one. A similar share of quarterly filers are

categorized as FCGs following the same definition as for monthly filers based on discrepancies in sales reported in the two returns. These potential FCGs react similarly to monthly filers but are less consequential for their downstream clients.

Column 1 of Table B.3 shows that quarterly filing FCGs also reduce their filing of credit-linked returns and Column 2 shows that they are unlikely to start making payments. Similar to their counterparts who are required to file at a monthly frequency, quarterly filer FCGs are more to become inactive (Column 3).

However, these quarterly filers are either more likely to be replaced or in other ways less consequential to their client firms. Just as with the monthly filers, we identify the client network of quarterly-filer FCGs as those who purchased intensively from these firms in the quarters before the blocking policy. We then compare their net tax payments and input tax credits used to that of a control group of firms with no purchases from quarterly-filer FCGs. Figure B.3 presents these event-study results. Unlike the network of monthly-filer FCGs, we see no measurable change in the net tax payments or input tax credit usage of these network firms after September 2021. The revenue effects of the GSTR-1 blocking policy among quarterly filers are therefore smaller.

Table B.1: Effect on Filing Behavior of Fake Credit Generators (Including Deregistered Firms)

	(1)	(2)	(3)	(4)
	Filed Credit-Linked Form	Filed Payment-Linked Form	Inactive	Deregistered
Fake Credit Generator × Post	-0.062	-0.017	0.054	0.019
	(0.004)	(0.003)	(0.004)	(0.004)
Outcome Mean	0.800	0.786	0.165	0.113
Observations	5,911,922	5,911,922	5,911,922	5,911,922

Notes: This table presents the causal effect of the policy on filing outcomes of Fake Credit Generators. This table only requires the firms to be registered in at least one period between January-June 2021. If a firm deregisters, they—by definition—do not file any of their forms. Therefore, we impute their outcomes following deregistration as being equal to 0. We classify firms as Fake Credit Generators using the same procedure described in Section 3.1. Column 1 presents the effect on filing of the credit-linked form in a given month. Column 2 presents the effect on filing of the payment-linked form in a given month. Column 3 presents the effect on filing of the neither form in a given month. Column 4 presents the effect on being deregistered in any given month. This outcome is imputed as the last known value in months when we do not know their status. In each case, the causal effect is estimated using the two-way fixed effects approach we describe in Section 4. We use a matched control group, where we match on pre-treatment firm characteristics (age, size, nature of business, whether the firm reports inter-state sales, and the number of filed pre-period payment returns) using Coarsened Exact Matching (CEM). We exclude all data after February, 2022 to remove any effect of delayed filing on the estimated treatment effect. Standard errors are clustered at the firm level. The outcome mean is the average of the outcome for the treatment group before the policy was implemented.

Table B.2: Effect on Network of FCGs (Including Deregistered FCGs)

	(1)	(2)	(3)
	Net Tax Payments (Share)	Credit Used (Share)	Log Gross Payments
Network Firm × Post	0.015	-0.015	-0.027
	(0.002)	(0.002)	(0.010)
Outcome Mean	0.246	0.754	11.785
Observations	1,910,588	1,910,588	1,910,588

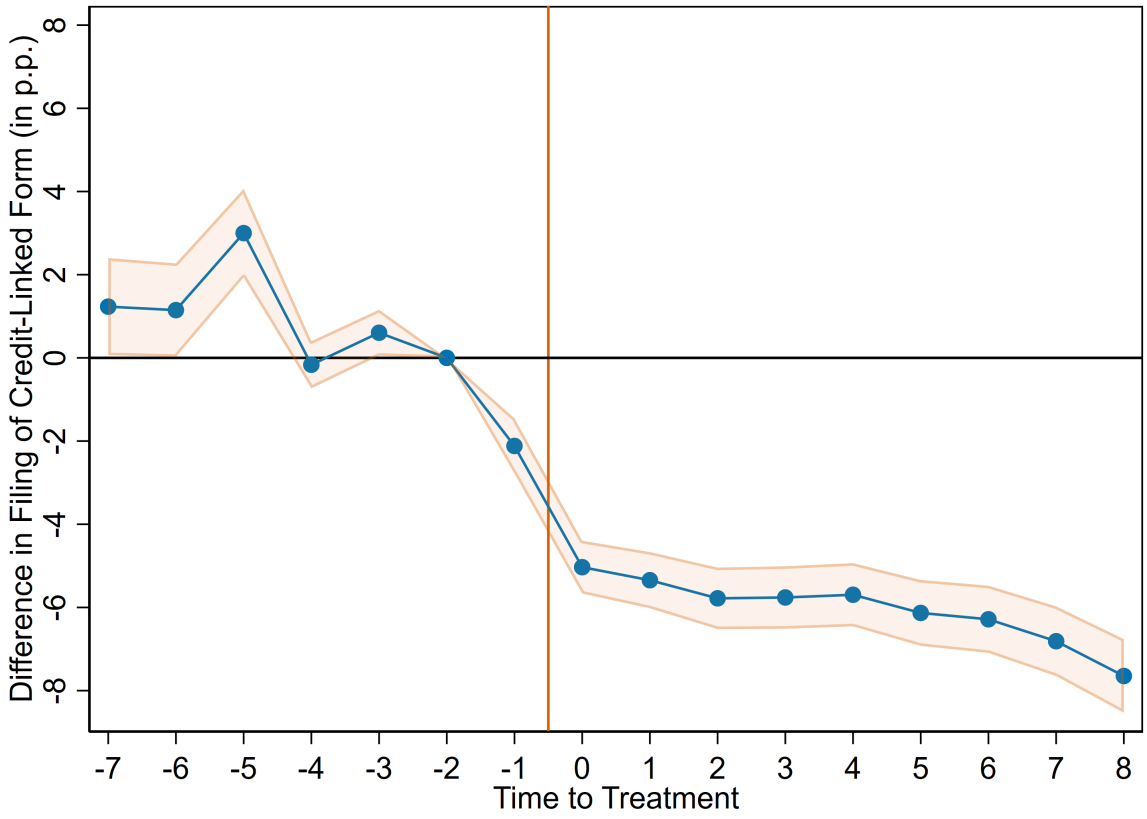
Notes: This table presents the causal effect of the policy on the network (buyers) of firms we classify as Fake Credit Generators. This table focuses on the network of the firms we classify as Fake Credit Generators that are registered in at least one period between January 2021-June 2021. The network firms themselves are registered in each period between January 2021-April 2022. We define the network of Fake Credit Generators as those firms who receive at least 5% of their total input tax credit between January-June 2021 from Fake Credit Generators. We compare these network firms to all firms who receive none of their input tax credits between January-June 2021 from Fake Credit Generators. We exclude all Fake Credit Generators themselves from this analysis. The outcome in Column 1 is defined as the ratio of net tax payments made by a firm using cash divided by the firm's total tax payments. The outcome in Column 2 is defined as the ratio of tax payments made by a firm using input tax credit divided by the firm's total tax payments. The outcome in Column 3 is defined as the log of the firm's gross tax payments in a given time period. We estimate the causal effects using the two-way fixed effects design described in Section 4. The outcome mean is the average of the outcome for the network firms before the policy was implemented. Standard errors are clustered at the firm-level.

Table B.3: Effect on Filing Behavior of Quarterly Fake Credit Generators (Including Deregistered Firms)

	(1)	(2)	(3)
	Filed Credit-Linked Form	Filed Payment-Linked Form	Inactive
Fake Credit Generator × Post	-0.032	-0.001	0.022
	(0.005)	(0.005)	(0.005)
Outcome Mean	0.871	0.853	0.121
Observations	1,557,749	1,557,749	1,557,749

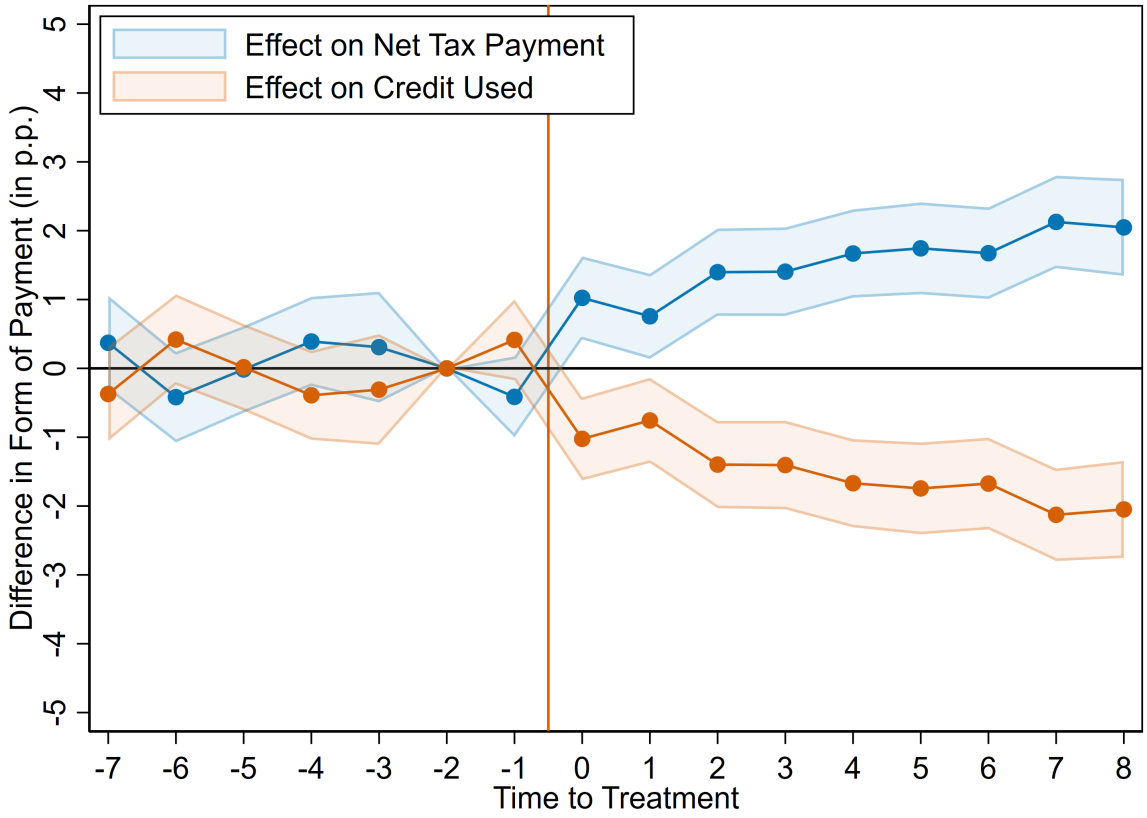
Notes: This table presents the causal effect of the policy on filing outcomes of Fake Credit Generators who elect to file their forms at a quarterly frequency. Each observation reflects the firm's filing outcome in that quarter. This table only requires the firms to be registered in at least one period between January-June 2021. If a firm deregisters, they—by definition—do not file any of their forms. Therefore, we impute their outcomes following deregistration as being equal to 0. We classify firms as Fake Credit Generators using the same procedure as in Section 3.1. Column 1 presents the effect on filing of the credit-linked form in a given quarter. Column 2 presents the effect on filing of the payment-linked form in a given quarter. Column 3 presents the effect on filing of neither form in a given quarter. In each case, the causal effect is estimated using the two-way fixed effects approach we describe in Section 4. We use a matched control group, where we match on pre-treatment firm characteristics (age, size, nature of business, whether the firm reports inter-state sales, and the number of filed pre-period payment returns) using Coarsened Exact Matching (CEM). We exclude all data after Q4 2021 to remove any effect of delayed filing on the estimated treatment effect. Standard errors are clustered at the firm level. The outcome mean is the average of the outcome for the treatment group before the policy was implemented.

Figure B.1: Effect of Policy on Filing of Credit-Linked Return (Including Deregistered Firms)



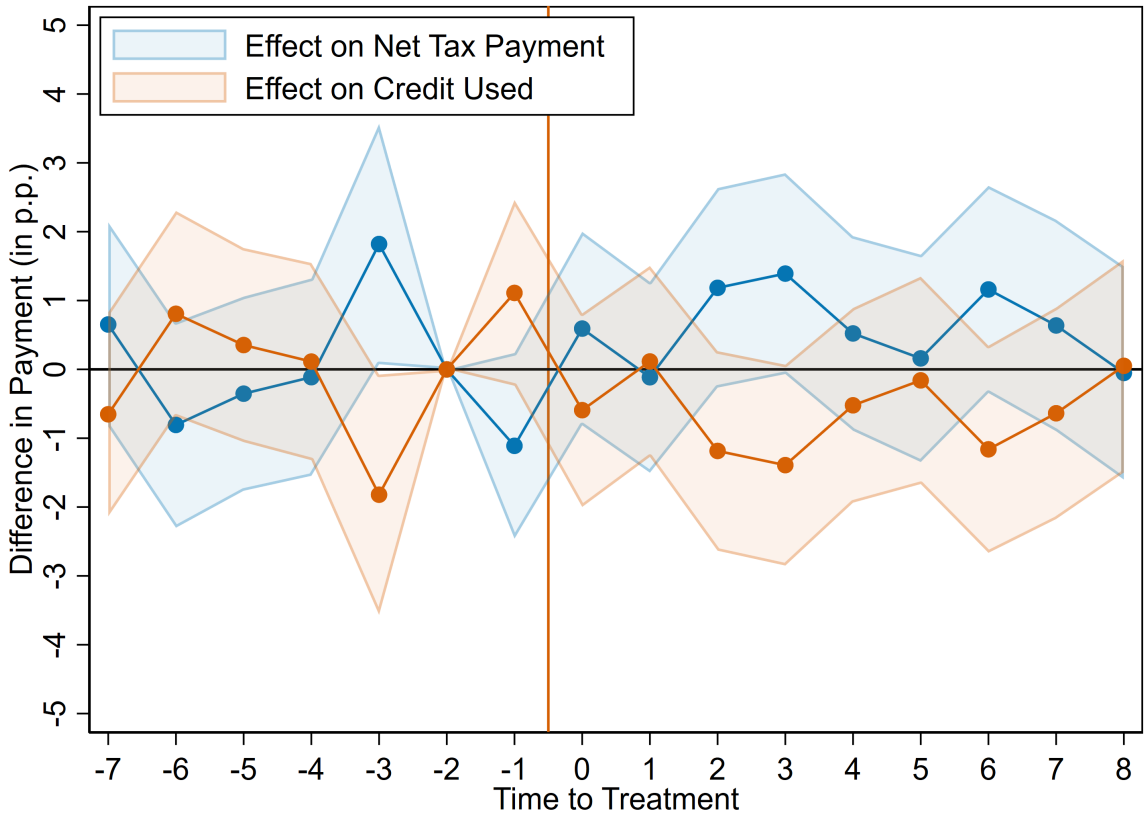
Notes. This figure shows the causal effect of the policy on filing of the credit-linked return by Fake Credit Generators. The sample includes firms that were registered in at least one month between January and June 2021. Once registered, a firm stays in the sample even if it is deregistered. Since deregistered firms do not file the credit-linked return by definition, the outcome is set to 0 for these firms. Firms are classified as Fake Credit Generators using the same approach described in Section 3.1. We plot the event-study coefficients from Equation 1 described in Section 4, along with the 95% confidence intervals. We use a matched control group, where we match on pre-treatment firm characteristics (age, size, nature of business, whether the firm reports inter-state sales, and the number of filed pre-period payment returns) using Coarsened Exact Matching (CEM). Filing outcomes in the periods 7 and 8 are affected by delayed filing and truncation of our data extracted in June 2022. Standard errors are clustered at the firm-level.

Figure B.2: Effect of Policy on Tax Payments by Network of FCGs (Including Deregistered FCGs)



Notes. This figure focuses on the network of the firms we classify as Fake Credit Generators that are registered in at least one period between January 2021-June 2021. The network firms themselves are registered in each period between January 2021-April 2022. We define the network of these FCGs as those firms who received at least 5% of their total input tax credit between January-June 2021 from FCGs and were themselves registered in at least one month between January and June 2021. We compare these network firms to all firms who received none of their input tax credits between January-June 2021 from FCGs and were registered in all periods between January 2021 - April 2022. We exclude all FCGs themselves from this analysis. The outcome is defined as the ratio of tax payments made using cash or input tax credit to the total tax payments in that return period. For a given firm, this ratio must necessarily sum to one in a given return period. We plot the event-study coefficients from Equation 1 (defining the network of FCGs as the treated firms) described in Section 4, along with the 95% confidence intervals. Standard errors are clustered at the firm-level.

Figure B.3: Effect on Tax Payments by Network of Quarterly Filer FCGs (Including Deregistered FCGs)



Notes. This figure focuses on the network of the firms we classify as Fake Credit Generators that are registered in at least one period between January 2021-June 2021 and elect to file their returns at a quarterly frequency. The network firms themselves are required to file their returns at a monthly frequency. We define the network of these FCGs as those firms who received at least 5% of their total input tax credit between January-June 2021 from FCGs and were themselves registered in at least one month between January and June 2021. We compare these network firms to all firms who received none of their input tax credits between January-June 2021 from FCGs and were registered in all periods between January 2021 - April 2022. We exclude all FCGs themselves from this analysis. The outcome is defined as the ratio of tax payments made using cash or input tax credit to the total tax payments in that return period. For a given firm, this ratio must necessarily sum to one in a given return period. We plot the event-study coefficients from Equation 1 (defining the network of quarterly filer FCGs as the treated firms) described in Section 4, along with the 95% confidence intervals. Standard errors are clustered at the firm-level.

C Revenue Calculations

In this appendix, we provide details of the methodology used to estimate the revenue implications of the blocking policy in Section 5.3. We measure the revenue implications by using estimates of two distinct effects of the policy: reduction in provision of fake credits by FCGs and increase in net tax payments by the clients of the FCGs. In subsections C.1-C.3, we show estimates of revenue impact using the estimated decline in credits issued and in payments made across different samples of FCGs, while in subsections C.4 and C.5, we estimate the revenue impact using the estimated increase in net tax payments of clients of FCGs. For our main sample, we construct 95% confidence intervals using a firm-level Bayesian bootstrap (Rubin, 1981) with 100 replications, implemented with Stata's bayesboot command.

C.1 Net Revenue Gain from FCGs: Main Sample (INR 1.5 billion)

We found in the main analysis that the FCGs significantly reduce their filing of credit-linked return in response to the blocking policy. By using the pre-reform average of the fake credits issued by these FCGs, we can estimate the decrease in fake credits because of the decrease in filing of credit-linked return.

Specifically, we had identified 5,811 firms in the pre-reform period as FCGs. The average per-firm credits issued by these firms was INR 4.7 million from January-June, 2021. This translates to a total of INR 4.52 billion monthly credit issued by all the FCGs in the pre-reform period. The estimated decline in filing of credit-linked return was 0.029, as reported in Column 1 of Table 1. This implies that the policy led to an annual decline in fraudulent credit of INR 1.56 billion. The 95% bootstrap confidence interval includes values between 0.83-2.29 billion.

The decline in credit overstates the revenue gain from FCGs, because FCGs also become 0.7 percentage points less likely to file their payment-linked return (Column 2 of Table 1) and therefore remit less tax. The average per-firm net payments by FCGs, i.e., tax payments made in cash rather than with input tax credit, were INR 0.77 million from January-June, 2021, or INR 0.75 billion per month across all FCGs. Applying the decline in filing of payment-linked return to this amount implies an annual decline in payments by FCGs of INR 0.06 billion. Netting this against the decline in credit, the annual revenue gain from FCGs is INR 1.50 billion. The 95% bootstrap confidence interval includes values between 0.80-2.19 billion.

$$
\begin{aligned}
\text{Monthly pre-policy credit issued by FCGs} &= \text{Mean credit issued per FCG, Jan–Jun 2021} \\
&\quad \times \text{Number of FCGs} \div 6 \\
&= \text{INR } 4{,}671{,}994 \times 5{,}811 \div 6 \\
&= \text{INR } 4.52 \text{ bn}
\end{aligned}
$$

$$\begin{aligned}\text{Annual decline in credit from FCGs} &= 12 \\ &\quad \times (-\text{ Effect on filing of credit-linked return}) \\ &\quad \times \text{Monthly pre-policy credit issued by FCGs} \\ &= 12 \times 0.0288 \times \text{INR } 4.52 \text{ bn} \\ &= \text{INR } 1.56 \text{ bn}\end{aligned}$$

$$\begin{aligned}\text{Monthly pre-policy net payments by FCGs} &= \text{Mean net payments per FCG,} \\ &\quad \text{Jan–Jun 2021} \\ &\quad \times \text{Number of FCGs} \div 6 \\ &= \text{INR } 774{,}941 \times 5{,}811 \div 6 \\ &= \text{INR } 0.75 \text{ bn}\end{aligned}$$

$$\begin{aligned}\text{Annual decline in payments by FCGs} &= 12 \\ &\quad \times (-\text{ Effect on filing of payment-linked return}) \\ &\quad \times \text{Monthly pre-policy net payments by FCGs} \\ &= 12 \times 0.0071 \times \text{INR } 0.75 \text{ bn} \\ &= \text{INR } 0.06 \text{ bn}\end{aligned}$$

$$\begin{aligned}\text{Annual net revenue gain from FCGs} &= \text{Annual decline in credit from FCGs} \\ &\quad - \text{Annual decline in payments by FCGs} \\ &= \text{INR } 1.56 \text{ bn} - \text{INR } 0.06 \text{ bn} \\ &= \text{INR } 1.50 \text{ bn}\end{aligned}$$

Table C.1: Inputs: Net Revenue Gain from FCGs, Main Sample

Input	Value	Source
Effect on filing of credit-linked return	−0.0288	Table 1, Panel C, Col. 1
Effect on filing of payment-linked return	−0.0071	Table 1, Panel C, Col. 2
Number of FCGs	5,811	Table A.1, Col. 2
Mean credit issued per FCG, Jan–Jun 2021	INR 4,671,994	Table A.1, Col. 2
Mean net payments per FCG, Jan–Jun 2021	INR 774,941	Table A.1, Col. 2

Notes: All inputs are computed on the main sample of monthly filers registered in every return period from January 2021 to April 2022. "Effect on filing of credit-linked return" and "Effect on filing of payment-linked return" are the coefficients on FCG × Post from two-way fixed effects regressions of, respectively, an indicator for filing the credit-linked return and an indicator for filing the payment-linked return in a given month on the interaction of an FCG indicator and a post-policy indicator (return periods August 2021 onward), with firm and return-period fixed effects, weighted by the Coarsened Exact Matching weights described in Section 4, and estimated on return periods January 2021 to February 2022. "Number of FCGs" is the count of firms in this sample classified as Fake Credit Generators using the procedure in Section 3.1. "Mean credit issued per FCG" is computed by summing, for each FCG, the input tax credit it reported on its credit-linked returns for the return periods January to June 2021, and averaging this firm-level total across the 5,811 FCGs; it corresponds to "Credit Issued" in Table A.1. "Mean net payments per FCG" is computed by summing, for each FCG, the tax payments it made in cash (as opposed to using input tax credit) on the payment-linked return for the return periods January to June 2021, and averaging this firm-level total across the 5,811 FCGs; it corresponds to "Net Payments" in Table A.1.

C.2 Net Revenue Gain from FCGs: Including Deregistered FCGs (INR 5.2 billion)

In the main analysis, we had only included firms that were registered for all return periods from January 2021 to April 2022 to ensure that the results are not affected by tax authority-driven composition changes. In Appendix B, we expand the sample by including FCGs that were registered for at least one return period in the pre-reform period. This almost doubles the number of FCGs from 5,811 in the main sample to 11,380 FCGs. Using this expanded sample in the revenue estimation increases the revenue impact of the policy by almost 3.5 times.

Although the average per-firm credits issued by FCGs in the expanded sample decreases to INR 3.8 million as compared to INR 4.7 million in the main sample, the total monthly credit issued in the pre-reform period in the expanded sample increases because of the larger number of FCGs. The estimated decrease in the filing of credit-linked return also approximately doubles from 2.9 percentage points to 6.2 percentage points. This increases the magnitude of annual decline in fraudulent credit to INR 5.39 billion, as compared to a decline of INR 1.56 billion in the main sample.

As in the main sample, we net out the decline in payments by FCGs. In the expanded sample, FCGs become 1.7 percentage points less likely to file their payment-linked return (Column 2 of Appendix Table B.1). The average per-firm net payments by FCGs were INR 0.44 million from January-June, 2021, or INR 0.84 billion per month across all FCGs, so the policy reduces annual payments by FCGs by INR 0.17 billion. The annual net revenue gain from FCGs in the expanded sample is therefore INR 5.22 billion.

$$\begin{aligned}\text{Monthly pre-policy credit issued by FCGs} &= \text{Mean credit issued per FCG, Jan–Jun 2021} \\ &\quad \times \text{Number of FCGs} \div 6 \\ &= \text{INR } 3{,}804{,}726 \times 11{,}380 \div 6 \\ &= \text{INR } 7.22 \text{ bn}\end{aligned}$$

$$\begin{aligned}\text{Annual decline in credit from FCGs} &= 12 \\ &\quad \times \big(- \text{ Effect on filing of credit-linked return}\big) \\ &\quad \times \text{Monthly pre-policy credit issued by FCGs} \\ &= 12 \times 0.0623 \times \text{INR } 7.22 \text{ bn} \\ &= \text{INR } 5.39 \text{ bn}\end{aligned}$$

$$\begin{aligned}\text{Monthly pre-policy net payments by FCGs} &= \text{Mean net payments per FCG,} \\ &\quad \text{Jan–Jun 2021} \\ &\quad \times \text{Number of FCGs} \div 6 \\ &= \text{INR } 443{,}883 \times 11{,}380 \div 6 \\ &= \text{INR } 0.84 \text{ bn}\end{aligned}$$

$$\begin{aligned}\text{Annual decline in payments by FCGs} &= 12 \\ &\quad \times \left(-\ \text{Effect on filing of payment-linked return}\right) \\ &\quad \times \text{Monthly pre-policy net payments by FCGs} \\ &= 12 \times 0.0172 \times \text{INR } 0.84 \text{ bn} \\ &= \text{INR } 0.17 \text{ bn}\end{aligned}$$

$$\begin{aligned}\text{Annual net revenue gain from FCGs} &= \text{Annual decline in credit from FCGs} \\ &\quad - \text{Annual decline in payments by FCGs} \\ &= \text{INR } 5.39 \text{ bn} - \text{INR } 0.17 \text{ bn} \\ &= \text{INR } 5.22 \text{ bn}\end{aligned}$$

Table C.2: Inputs: Net Revenue Gain from FCGs, Including Deregistered FCGs

Input	Value	Source
Effect on filing of credit-linked return	−0.0623	Table B.1, Col. 1
Effect on filing of payment-linked return	−0.0172	Table B.1, Col. 2
Number of FCGs	11,380	-
Mean credit issued per FCG, Jan–Jun 2021	INR 3,804,726	-
Mean net payments per FCG, Jan–Jun 2021	INR 443,883	-

Notes: All inputs are computed on the expanded sample of monthly filers registered for at least one return period between January and June 2021 (Appendix B). Once registered, a firm remains in the sample in every subsequent return period, and its filing outcomes are set to zero in periods in which it files no return, including after deregistration. "Effect on filing of credit-linked return" and "Effect on filing of payment-linked return" are the coefficients on FCG × Post from two-way fixed effects regressions of, respectively, an indicator for filing the credit-linked return and an indicator for filing the payment-linked return in a given month on the interaction of an FCG indicator and a post-policy indicator (return periods August 2021 onward), with firm and return-period fixed effects, weighted by Coarsened Exact Matching weights constructed for this sample, and estimated on return periods January 2021 to February 2022. "Number of FCGs" is the count of firms in this sample classified as Fake Credit Generators using the procedure in Section 3.1. "Mean credit issued per FCG" is computed by summing, for each FCG, the input tax credit it reported on its credit-linked returns for the return periods January to June 2021, and averaging this firm-level total across the 11,380 FCGs. "Mean net payments per FCG" is computed by summing, for each FCG, the tax payments it made in cash (as opposed to using input tax credit) on the payment-linked returns for the return periods January to June 2021, and averaging this firm-level total across the 11,380 FCGs.

C.3 Net Revenue Gain from Quarterly-Filer FCGs (INR 0.1 billion)

Only firms with annual revenue below INR 50 million can file on a quarterly basis. We find that the revenue impact of decline in filing of credit-linked returns by FCGs filing on quarterly basis is small. The average per-firm credits issued by quarterly-filer FCGs is INR 350,000, much lower than for FCGs that file on a monthly basis. This implies that the total quarterly credits issued by 4,856 quarterly-filing FCGs in the pre-reform period is only INR 0.85 billion. Even though the blocking policy leads to a 3.24 percentage points decline in the filing of credit-linked return in this group, the total decline in fraudulent credits is only INR 0.11 billion. Quarterly-filer FCGs show no meaningful change in filing of their payment-linked return (Column 2 of Table B.3), and their pre-reform net payments are small (INR 0.09 billion per quarter), so the decline in payments is negligible (INR 0.0003 billion) and the annual net revenue gain from this group remains INR 0.11 billion.

$$\begin{aligned}\text{Quarterly pre-policy credit issued by FCGs} &= \text{Mean credit issued per FCG, Jan–Jun 2021} \\ &\quad \times \text{Number of FCGs} \div 2 \\ &= \text{INR } 350{,}038 \times 4{,}856 \div 2 \\ &= \text{INR } 0.85 \text{ bn}\end{aligned}$$

$$\begin{aligned}\text{Annual decline in credit from FCGs} &= 4 \\ &\quad \times \big(-\ \text{Effect on filing of credit-linked return}\big) \\ &\quad \times \text{Quarterly pre-policy credit issued by FCGs} \\ &= 4 \times 0.0324 \times \text{INR } 0.85 \text{ bn} \\ &= \text{INR } 0.11 \text{ bn}\end{aligned}$$

$$\begin{aligned}\text{Quarterly pre-policy net payments by FCGs} &= \text{Mean net payments per FCG,} \\ &\quad \text{Jan–Jun 2021} \\ &\quad \times \text{Number of FCGs} \div 2 \\ &= \text{INR } 38{,}071 \times 4{,}856 \div 2 \\ &= \text{INR } 0.09 \text{ bn}\end{aligned}$$

$$
\begin{aligned}
\text{Annual decline in payments by FCGs} &= 4 \\
&\quad \times (-\text{ Effect on filing of payment-linked return}) \\
&\quad \times \text{Quarterly pre-policy net payments by FCGs} \\
&= 4 \times 0.0008 \times \text{INR } 0.09 \text{ bn} \\
&= \text{INR } 0.0003 \text{ bn}
\end{aligned}
$$

$$
\begin{aligned}
\text{Annual net revenue gain from FCGs} &= \text{Annual decline in credit from FCGs} \\
&\quad - \text{Annual decline in payments by FCGs} \\
&= \text{INR } 0.11 \text{ bn} - \text{INR } 0.0003 \text{ bn} \\
&= \text{INR } 0.11 \text{ bn}
\end{aligned}
$$

Table C.3: Inputs: Net Revenue Gain from Quarterly-Filer FCGs

Input	Value	Source
Effect on filing of credit-linked return	−0.0324	Table B.3, Col. 1
Effect on filing of payment-linked return	−0.0008	Table B.3, Col. 2
Number of FCGs	4,856	-
Mean credit issued per FCG, Jan–Jun 2021	INR 350,038	-
Mean net payments per FCG, Jan–Jun 2021	INR 38,071	-

Notes: All inputs are computed on the sample of firms that elect to file their returns quarterly and were registered for at least one return period between January and June 2021. Each observation is a firm-quarter, and outcomes are set to zero in quarters in which the firm files no return. "Effect on filing of credit-linked return" and "Effect on filing of payment-linked return" are the coefficients on FCG × Post from two-way fixed effects regressions of, respectively, an indicator for filing the credit-linked return and an indicator for filing the payment-linked return in a given quarter on the interaction of an FCG indicator and a post-policy indicator (return quarters from the third quarter of 2021 onward), with firm and return-quarter fixed effects, weighted by Coarsened Exact Matching weights constructed for this sample, and estimated on the four return quarters of 2021. "Number of FCGs" is the count of quarterly filers classified as Fake Credit Generators using the procedure in Section 3.1 applied to their January–June 2021 returns. "Mean credit issued per FCG" is computed by summing, for each FCG, the input tax credit it reported on its credit-linked returns for the two return quarters covering January to June 2021, and averaging this firm-level total across the 4,856 FCGs. "Mean net payments per FCG" is computed by summing, for each FCG, the tax payments it made in cash (as opposed to using input tax credit) on the payment-linked returns for the two return quarters covering January to June 2021, and averaging this firm-level total across the 4,856 FCGs.

C.4 Increase in Net Tax Payments by Client Firms: Main Sample (INR 2.2 billion)

In this subsection, we use the responses of clients of FCGs to estimate the revenue impact of the policy. A reduction in the use of fake input tax credits increases government revenue by shifting firms from paying their tax liability with input tax credits towards making cash payments. However, the overall revenue effect also depends on firms' total reported tax liability after the policy change. Column 1 of Table 2 shows that clients of FCGs increase their share of tax payments paid in cash by 1.38 percentage points. This increases the total share of cash payments from 25.93 percent to 27.31 percent. At the same time, their reported gross tax liability declines by 1.75 percent, from INR 20.60 billion per month to INR 20.24 billion per month. Accounting for both effects—the increase in the cash share and the decline in reported gross tax liability—we estimate that the policy generates an annual revenue gain of INR 2.24 billion. The 95% bootstrap confidence interval includes values between 0.66-3.82 billion.

$$
\begin{aligned}
\text{Pre-policy monthly gross payments} &= \text{Mean gross payments per network firm, Jan–Jun 2021} \\
&\quad \times \text{Number of network firms} \div 6 \\
&= \text{INR } 17{,}727{,}650 \times 6{,}973 \div 6 \\
&= \text{INR } 20.60 \text{ bn}
\end{aligned}
$$

$$
\begin{aligned}
\text{Post-policy monthly gross payments} &\approx (1 + \text{Effect on log gross payments}) \\
&\quad \times \text{Pre-policy monthly gross payments} \\
&= (1 - 0.0175) \times \text{INR } 20.60 \text{ bn} \\
&= \text{INR } 20.24 \text{ bn}
\end{aligned}
$$

$$
\begin{aligned}
\text{Annual revenue gain} &= 12 \times \Big[(\text{Pre-policy cash share} + \text{Effect on cash share}) \\
&\quad \times \text{Post-policy monthly gross payments} \\
&\quad - \text{Pre-policy cash share} \times \text{Pre-policy monthly gross payments} \Big] \\
&= 12 \times \big[(0.2593 + 0.0138) \times \text{INR } 20.24 \text{ bn} - 0.2593 \times \text{INR } 20.60 \text{ bn} \big] \\
&= \text{INR } 2.24 \text{ bn}
\end{aligned}
$$

Table C.4: Inputs: Increase in Net Tax Payments by Client Firms, Main Sample

Input	Value	Source
Effect on log gross payments	−0.0175	Table 2, Panel C, Col. 3
Effect on cash share of payments	0.0138	Table 2, Panel C, Col. 1
Number of network firms	6,973	Table A.4, Col. 2
Mean gross payments per network firm, Jan–Jun 2021	INR 17,727,650	Table A.4, Col. 2
Pre-policy cash share of payments (ratio of net tax payments to gross payments)	0.2593	Table A.4, Col. 2

Notes: All inputs are computed on the main sample of monthly filers registered in every return period from January 2021 to April 2022, excluding FCGs themselves. Network firms are firms that received at least 5 percent of their total input tax credit between January and June 2021 from FCGs; the control group consists of firms that received no input tax credit from FCGs in that period, additionally excluding the clients of quarterly-filing FCGs. Gross tax payments are the firm's total tax payments reported on the payment-linked return in a return period, i.e., the sum of payments made in cash and payments made using input tax credit. "Effect on log gross payments" and "Effect on cash share of payments" are the coefficients on Network Firm × Post from unweighted two-way fixed effects regressions of, respectively, the log of gross tax payments and the ratio of cash payments to gross tax payments on the interaction of a network-firm indicator and a post-policy indicator (return periods August 2021 onward), with firm and return-period fixed effects, estimated on return periods January 2021 to April 2022. "Number of network firms" is the count of network firms. "Mean gross payments per network firm" is computed by summing, for each network firm, its gross tax payments over the return periods January to June 2021, and averaging this firm-level total across the 6,973 network firms; it corresponds to "Gross Payments" in Table A.4. "Pre-policy cash share of payments" is the ratio of cash payments to gross tax payments in a firm-month, averaged across all firm-months of network firms between January and June 2021; firm-months with zero gross tax payments are excluded. It corresponds to "Ratio of Net Tax Payments to Gross Payments" in Table A.4; the outcome mean in Column 1 of Table 2, Panel C, which additionally includes July 2021, is identical to three decimal places.

C.5 Increase in Net Tax Payments by Client Firms: Including Clients of Deregistered FCGs (INR 2.4 billion)

Column 1 of Table B.2 shows that clients of FCGs increase their share of tax payments paid in cash by 1.51 percentage points. This increases the total share of cash payments from 24.57 percent to 26.08 percent. At the same time, their reported gross tax liability declines by 2.74 percent, from INR 25.40 billion per month to INR 24.70 billion per month. Accounting for both effects–the increase in the cash share and the decline in reported gross tax liability–we estimate that the policy generates an annual revenue gain of INR 2.42 billion.

$$\begin{aligned}\text{Pre-policy monthly gross payments} &= \text{Mean gross payments per network firm, Jan–Jun 2021} \\ &\quad \times \text{Number of network firms} \div 6 \\ &= \text{INR } 13{,}600{,}226 \times 11{,}205 \div 6 \\ &= \text{INR } 25.40 \text{ bn}\end{aligned}$$

$$\begin{aligned}\text{Post-policy monthly gross payments} &\approx (1 + \text{Effect on log gross payments}) \\ &\quad \times \text{Pre-policy monthly gross payments} \\ &= (1 - 0.0274) \times \text{INR } 25.40 \text{ bn} \\ &= \text{INR } 24.70 \text{ bn}\end{aligned}$$

$$\begin{aligned}\text{Annual revenue gain} &= 12 \times \Big[(\text{Pre-policy cash share} + \text{Effect on cash share}) \\ &\quad \times \text{Post-policy monthly gross payments} \\ &\quad - \text{Pre-policy cash share} \times \text{Pre-policy monthly gross payments}\Big] \\ &= 12 \times \big[(0.2457 + 0.0151) \times \text{INR } 24.70 \text{ bn} - 0.2457 \times \text{INR } 25.40 \text{ bn}\big] \\ &= \text{INR } 2.42 \text{ bn}\end{aligned}$$

Table C.5: Inputs: Increase in Net Tax Payments by Client Firms, Including Clients of Deregistered FCGs

Input	Value	Source
Effect on log gross payments	−0.0274	Table B.2, Col. 3
Effect on cash share of payments	0.0151	Table B.2, Col. 1
Number of network firms	11,205	-
Mean gross payments per network firm, Jan–Jun 2021	INR 13,600,226	-
Pre-policy cash share of payments	0.2457	Table B.2, Col. 1, Outcome Mean

Notes: All inputs are computed on the network of the expanded sample of FCGs, i.e., FCGs registered for at least one return period between January and June 2021 (Appendix B). The network firms and control firms themselves are monthly filers registered in every return period from January 2021 to April 2022, excluding FCGs. Network firms are firms that received at least 5 percent of their total input tax credit between January and June 2021 from expanded-sample FCGs; the control group consists of firms that received no input tax credit from FCGs in that period, additionally excluding the clients of quarterly-filing FCGs. Gross tax payments are the firm's total tax payments reported on the payment-linked return in a return period, i.e., the sum of payments made in cash and payments made using input tax credit. "Effect on log gross payments" and "Effect on cash share of payments" are the coefficients on Network Firm × Post from unweighted two-way fixed effects regressions of, respectively, the log of gross tax payments and the ratio of cash payments to gross tax payments on the interaction of a network-firm indicator and a post-policy indicator (return periods August 2021 onward), with firm and return-period fixed effects, estimated on return periods January 2021 to April 2022. "Number of network firms" is the count of network firms. "Mean gross payments per network firm" is computed by summing, for each network firm, its gross tax payments over the return periods January to June 2021, and averaging this firm-level total across the 11,205 network firms. "Pre-policy cash share of payments" is the ratio of cash payments to gross tax payments in a firm-month, averaged across all firm-months of network firms between January and June 2021; firm-months with zero gross tax payments are excluded. The outcome mean in Column 1 of Table B.2, which additionally includes July 2021, is identical to three decimal places.